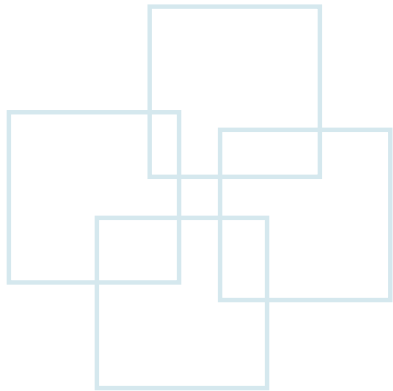




國立成功大學  
National Cheng Kung University

# Introduction to Emerging Memory Techniques

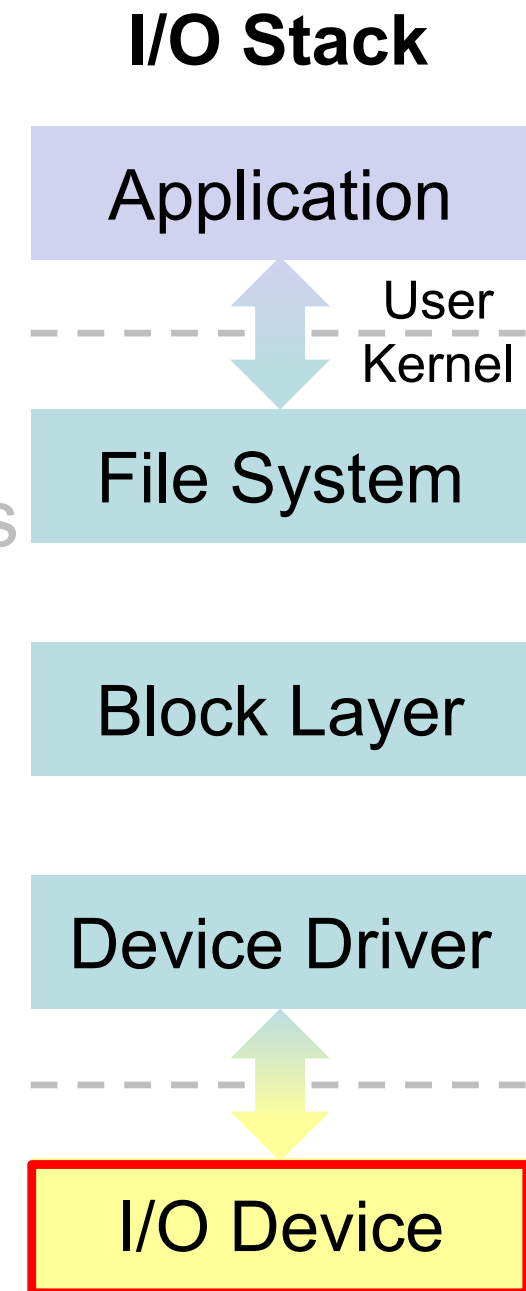
## Lecture 3: Shingled Magnetic Recording



Prof. Chien-Chung Ho

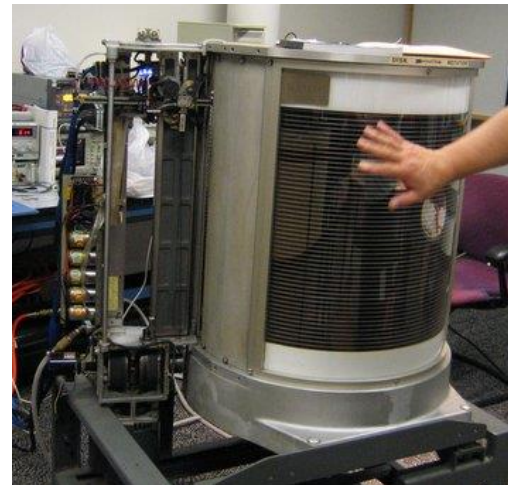
# Agenda

- Traditional Hard Disk Drive
  - Why and How
  - Development Bottleneck
- New Magnetic Recording Technologies
- Shingled Magnetic Recording (SMR)
  - Basics and Inherent Challenges
  - General Solution: Persistent Cache
  - Various SMR Drive Models and Designs
    - Drive-Managed SMR (DM-SMR)
    - Host-Aware SMR (HA-SMR)
    - Host-Managed SMR (HM-SMR)
    - Hybrid SMR



# History of Hard Disk Drives

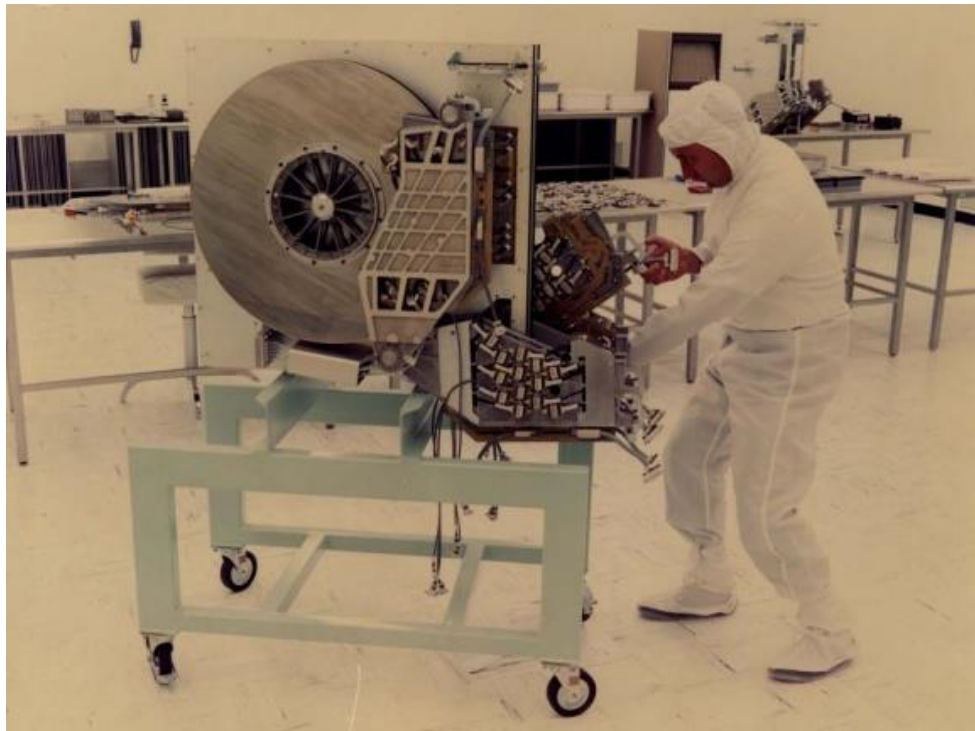
- HDDs have been the main form of persistent data storage in computer systems for decades.
  - In 1953, IBM recognized the urgent need.
  - The first commercial usage of HDD began in 1957.
    - Model 350 RAMAC actuator and disk stack.
    - 5 MB capacity, 0.0088MB/s write throughput.



- Many file and storage systems are designed and optimized based on HDD characteristics.

# Amazing Photos about HDDs

- Below is a **250MB** hard disk drive from 1979 ...
  - <https://isite.tw/2018/12/02/19503>
- HDDs were used together with big mainframe.
  - Entire rooms were already set aside for the computers.



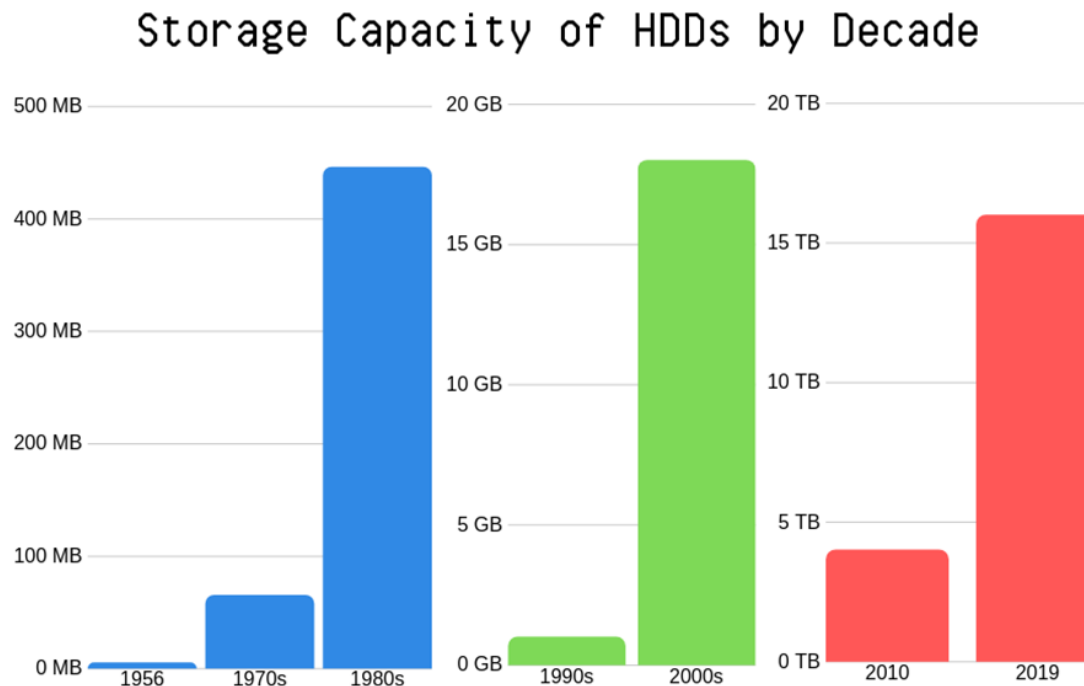
# Amazing Photos about HDDs

- From '80s to today: 8-inch → 3.5, 2.5, 1.8-inch drives



# Changes in Storage Space Over Time

- The first HDD could store **5MB** of data in 1956.
  - As same as size of the first **small** 5.25-inch HDDs that arrived in 1980.
- A “normal” HDD held about 40~100MB in 1990.
- Today, you can buy a 3.5-inch HDDs with 18TB.

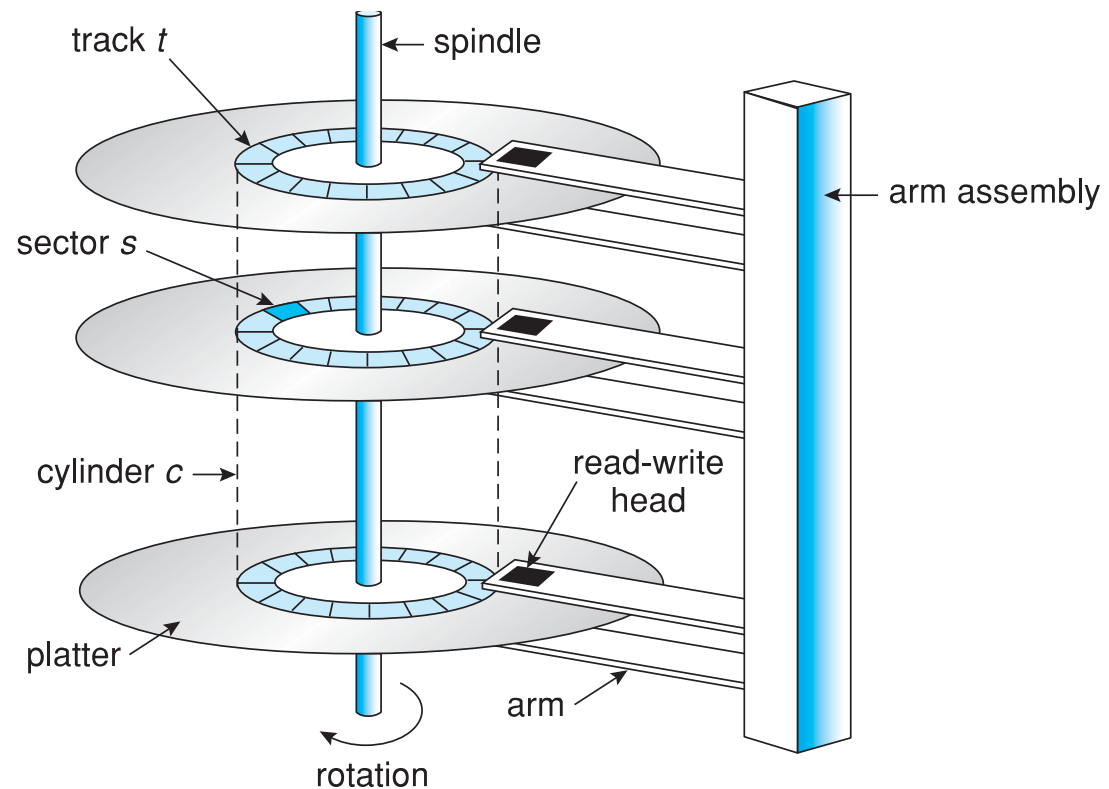


# Recall: Disk Organization: Logical View

- HDD: Accessed in **blocks** but organized in **sectors**
  - **Sector**
    - The most common sector size is 512 bytes.
      - The Sector size is fixed on an HDD
    - All sectors are numbered from 0 to  $n - 1$  (i.e., the **address space**).
      - The disk can be logically viewed as **an array of  $n$  sectors**.
  - **Block**
    - Disk I/Os are in units of **blocks**.
    - A **block** may refer to one or multiple sectors.
- In an HDD, only a single 512-byte write is **atomic**.
  - It will either complete in entirety or fail at all.
    - **Torn Write**: Only a portion of a larger write may complete.

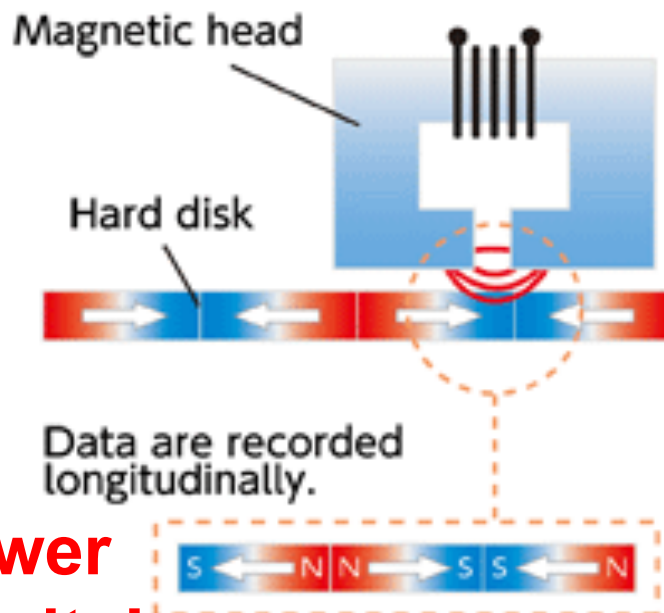
# Recall: Disk Organization: Physical View

- A hard disk has one or multiple platters.
  - Each platter has 2 sides (**surfaces**).
  - Platters are bound together by a spindle.
- Each surface has multiple concentric circles called **tracks**.
  - A track is further divided into **sectors**.
- A disk head reads or writes data of sectors.

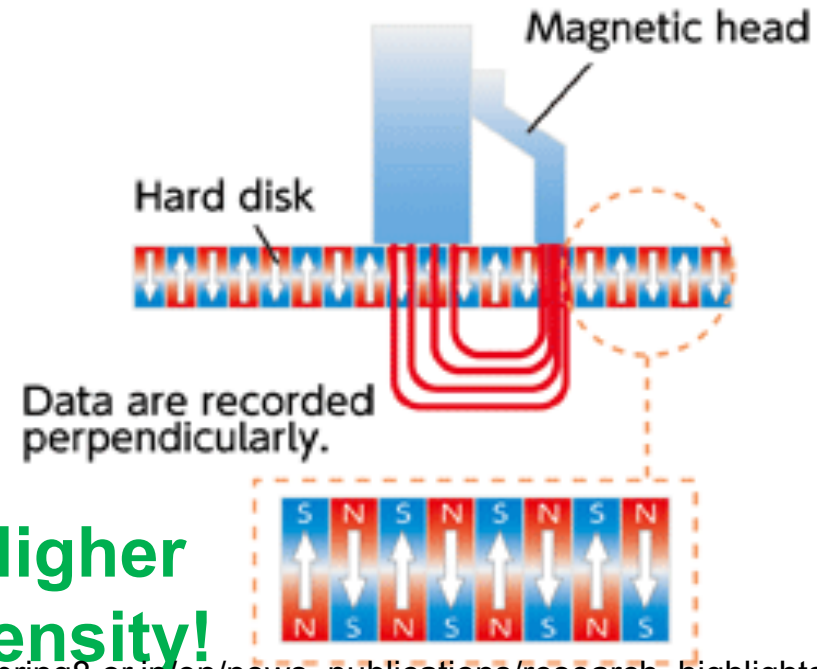


# Magnetic Recording Technology

- HDDs store the data as tiny area of either positive or negative **magnetization** on the disk surface.
  - Each tiny area represents a “**bit**” of information.
  - Based on the magnetization direction, there are two **Conventional Magnetic Recording (CMR)** technologies:
    - ① Longitudinal **Magnetic Recording**    ② Perpendicular **Magnetic Recording**



**Lower  
Density!**

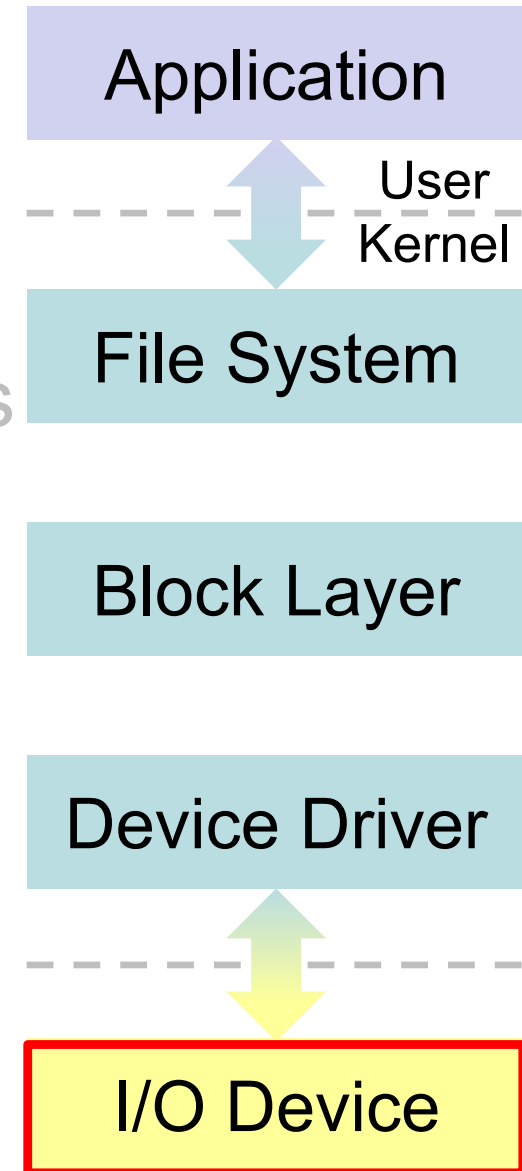


**Higher  
Density!**

# Agenda

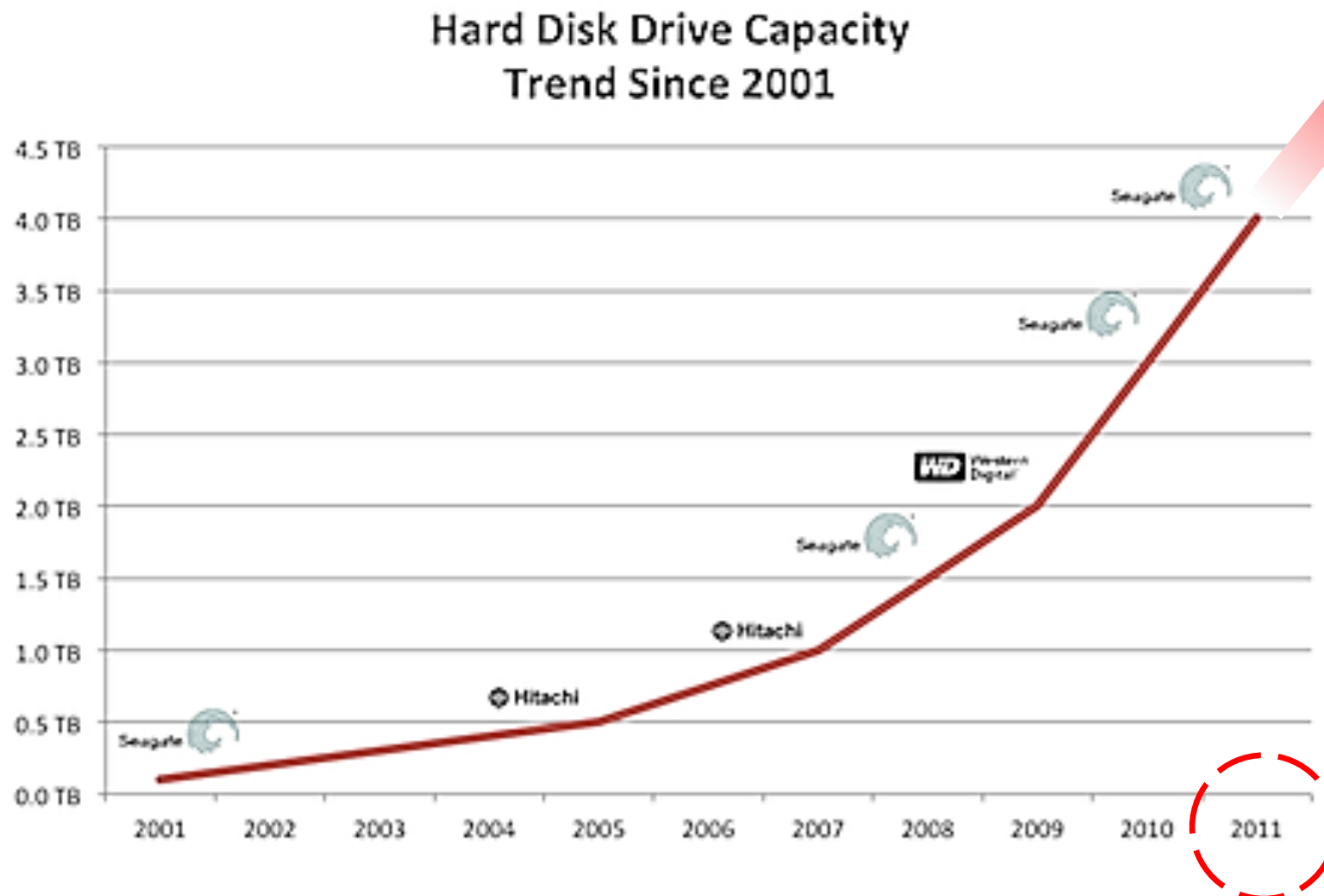
- Traditional Hard Disk Drive
  - Why and How
  - Development Bottleneck
- New Magnetic Recording Technologies
- Shingled Magnetic Recording (SMR)
  - Basics and Inherent Challenges
  - General Solution: Persistent Cache
  - Various SMR Drive Models and Designs
    - Drive-Managed SMR (DM-SMR)
    - Host-Aware SMR (HA-SMR)
    - Host-Managed SMR (HM-SMR)
    - Hybrid SMR

## I/O Stack



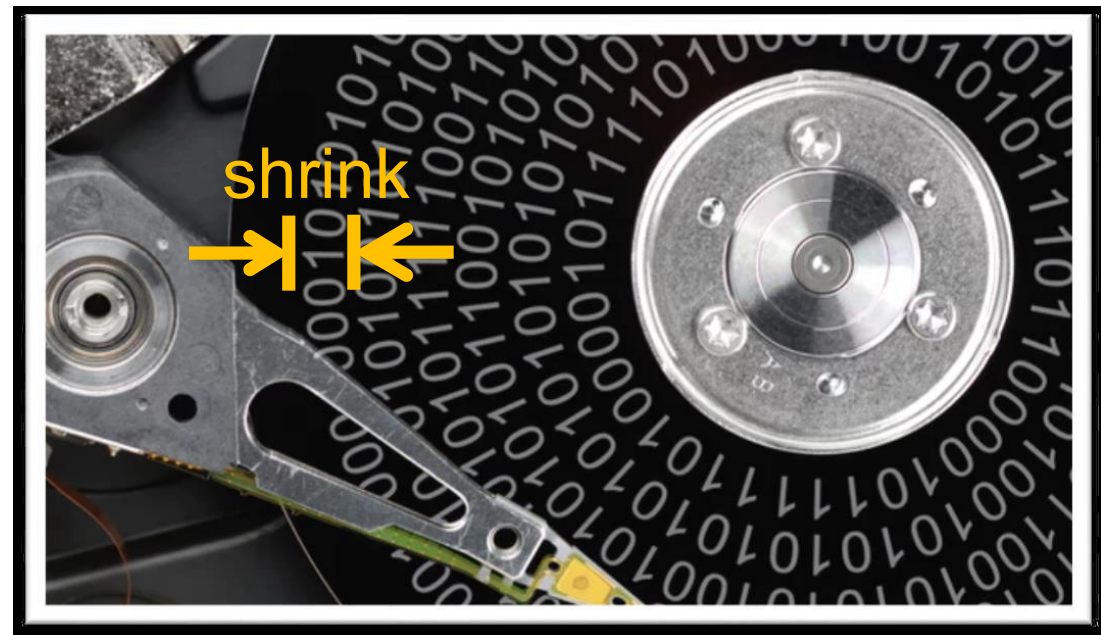
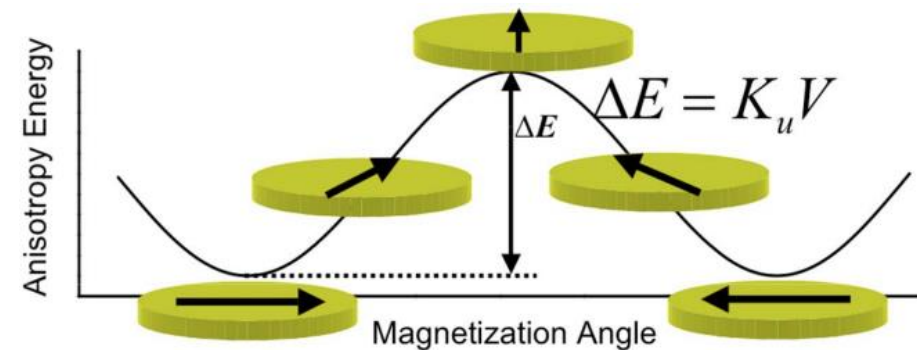
# HDD Capacity Trend

- The capacity of HDDs increased at **breakneck speed** in past years.



# Shrinkage Bottleneck

- PMR has reached the **bottleneck** in providing **higher areal density**.
  - The maximal areal density is **1 TB per square inch**.
  - Because of the **superparamagnetic effect (SPE)**, it is **hard** to continuously shrink the volume of magnetic grains.



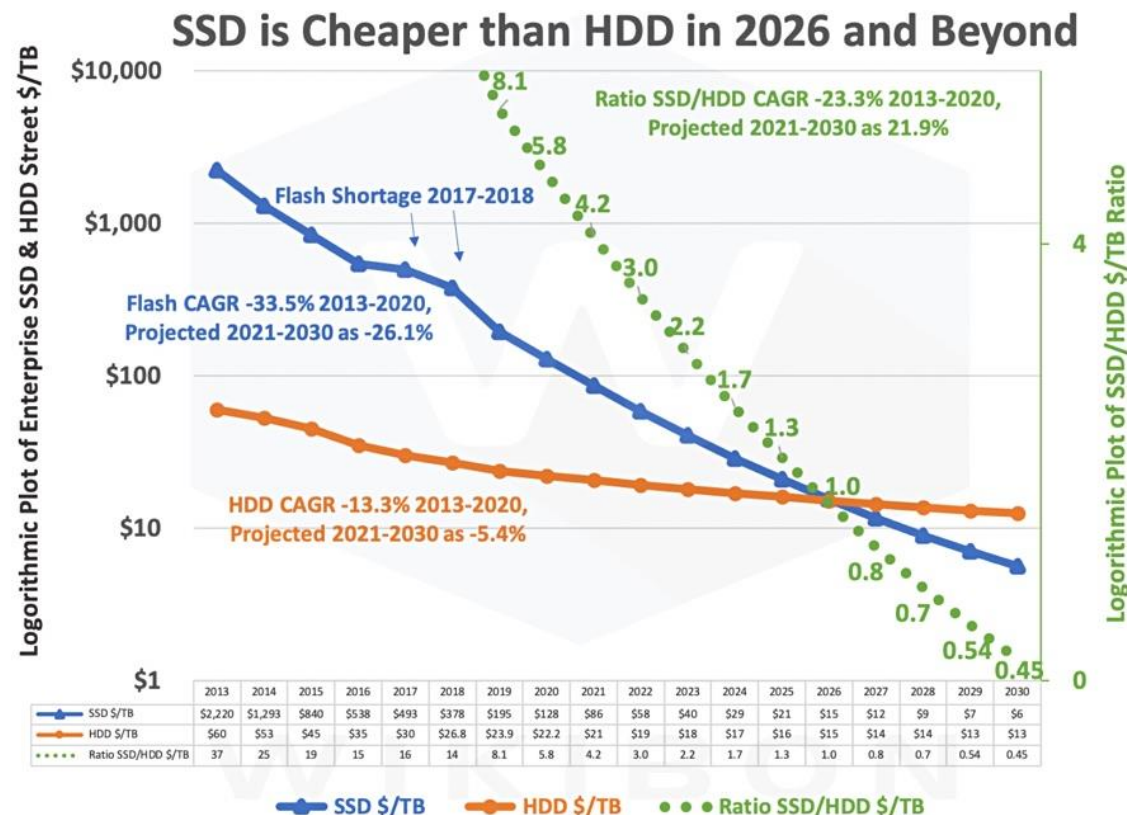
# Breaking the Bottleneck or Dying...

- **Strong Rival: Solid-State Drive (SSD)**
  - Flash memory demonstrates several good advantages ...
  - Flash memory is getting **cheaper** with **degraded reliability**.

## SSD



- ✓ **Faster**
- ✓ **Silent**
- ✓ **Shock-resistant**
- ✓ **Energy efficient**
- ✓ **Lighter**
- ✓ **Smaller**



Source: © Wikibon, 2021. The 2021-2030 Enterprise Street Prices are Projected by Wikibon using Wright's Law

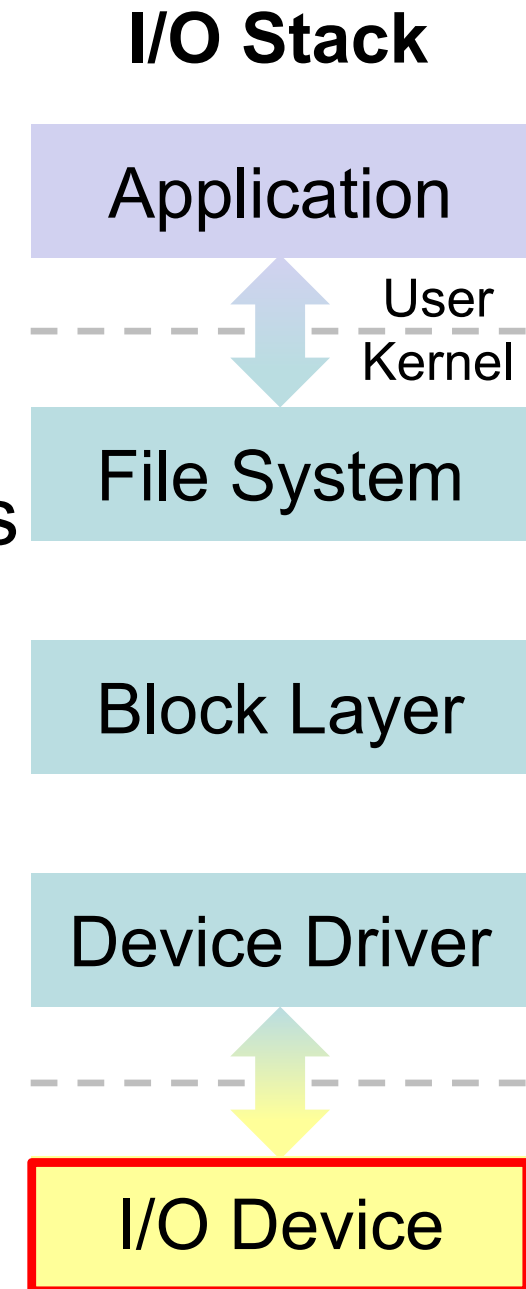
## HDD



- ✓ **Cheaper?**
- ✓ **Reliability**

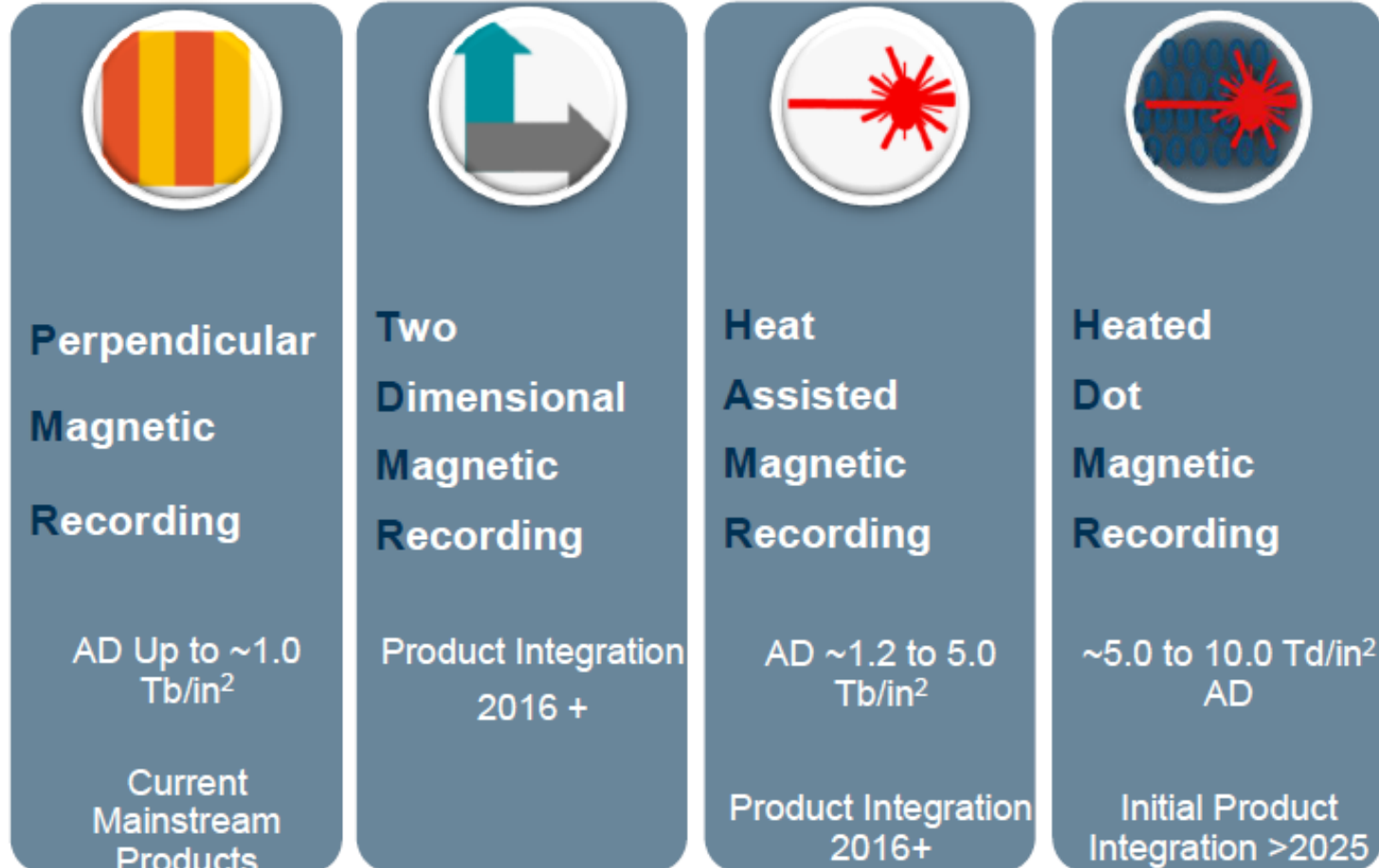
# Agenda

- Traditional Hard Disk Drive
  - Why and How
  - Development Bottleneck
- New Magnetic Recording Technologies
- Shingled Magnetic Recording (SMR)
  - Basics and Inherent Challenges
  - General Solution: Persistent Cache
  - Various SMR Drive Models and Designs
    - Drive-Managed SMR (DM-SMR)
    - Host-Aware SMR (HA-SMR)
    - Host-Managed SMR (HM-SMR)
    - Hybrid SMR



# New Magnetic Recording Technologies

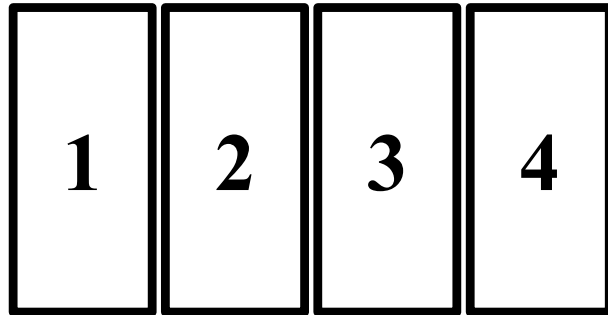
- HDDs are poised to keep evolving for the **increased areal density (AD)** with **various new technologies**:
  - **Less** things to do with firmware/software



# New Track Layouts

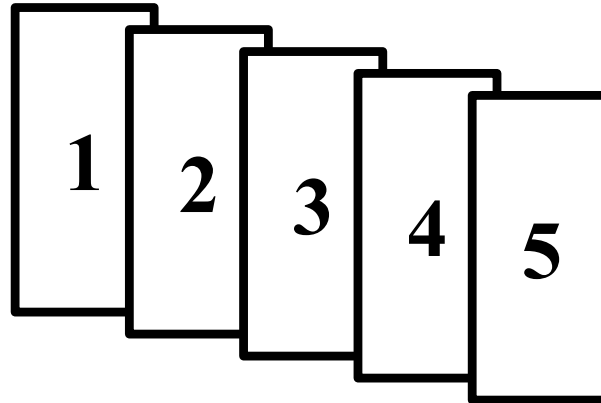
- HDDs are poised to keep evolving for the **increased areal density (AD)** with **different track layouts**:
  - **More** things to do with **firmware/software**

**Conventional**  
magnetic recording



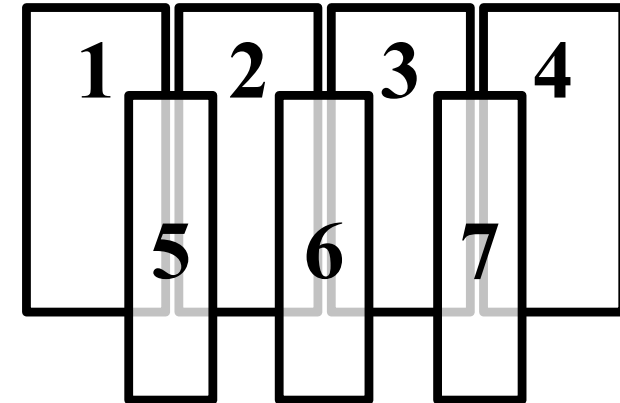
- ✓ Tracks **non-overlap**
  - Tracks can be **rewritten freely**

**Shingled**  
magnetic recording



- ✓ Tracks **overlap**
  - **25% higher capacity** than CMR
  - **Commercially available now**
  - **Track rewrite issue**

**Interlaced**  
magnetic recording

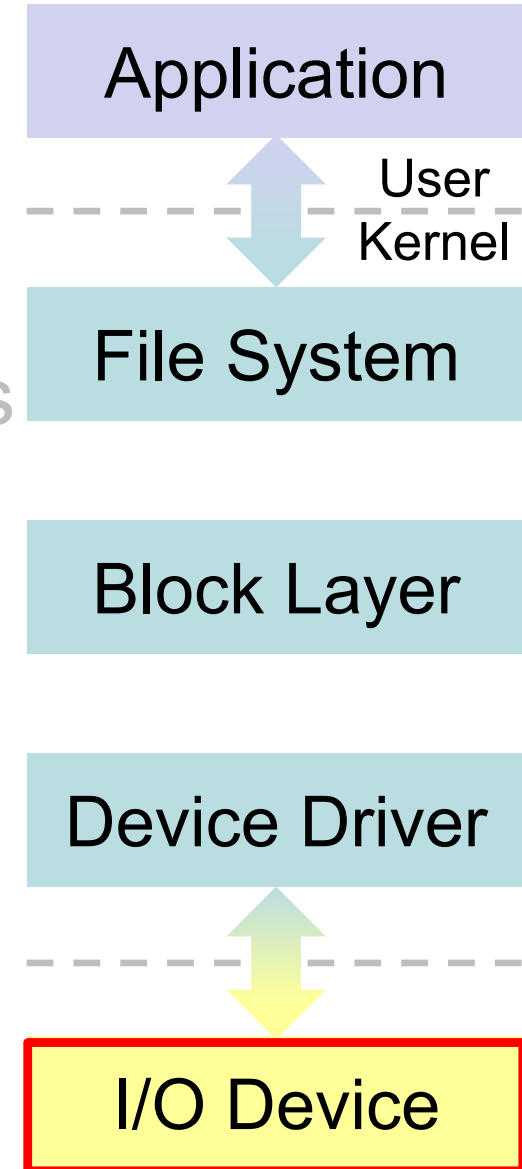


- ✓ Tracks **overlap**
  - **40% higher capacity** than CMR
  - **Not commercially available yet**
  - **Track rewrite issue**

# Agenda

- Traditional Hard Disk Drive
  - Why and How
  - Development Bottleneck
- New Magnetic Recording Technologies
- Shingled Magnetic Recording (SMR)
  - Basics and Inherent Challenges
  - General Solution: Persistent Cache
  - Various SMR Drive Models and Designs
    - Drive-Managed SMR (DM-SMR)
    - Host-Aware SMR (HA-SMR)
    - Host-Managed SMR (HM-SMR)
    - Hybrid SMR

## I/O Stack



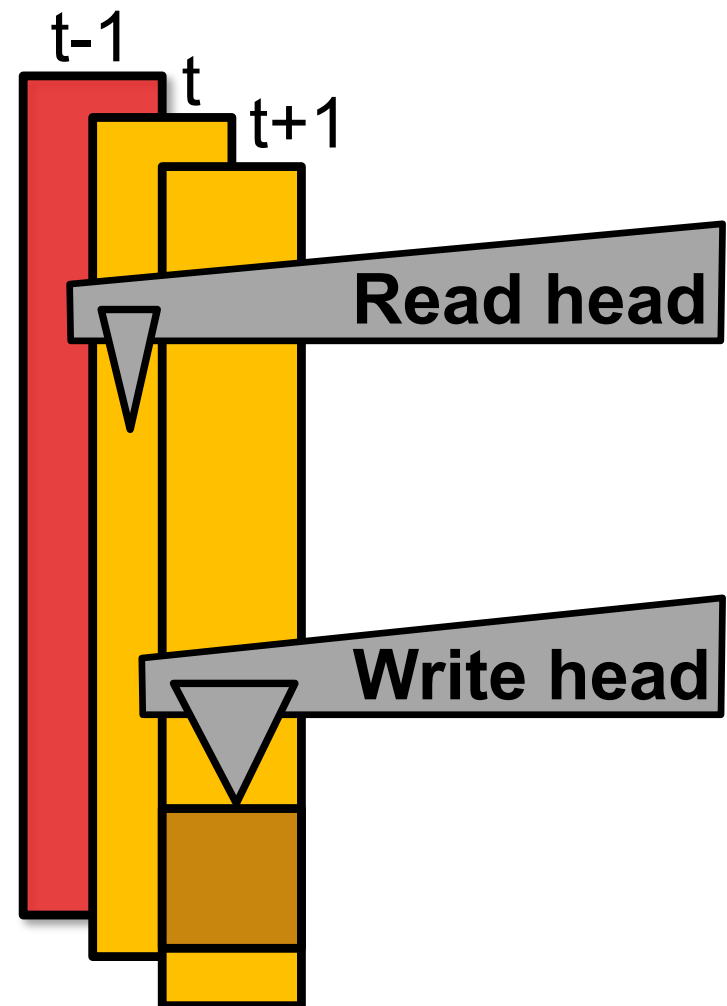
# Shingle?

- <https://www.youtube.com/watch?v=O8ZP-jlf8bs>



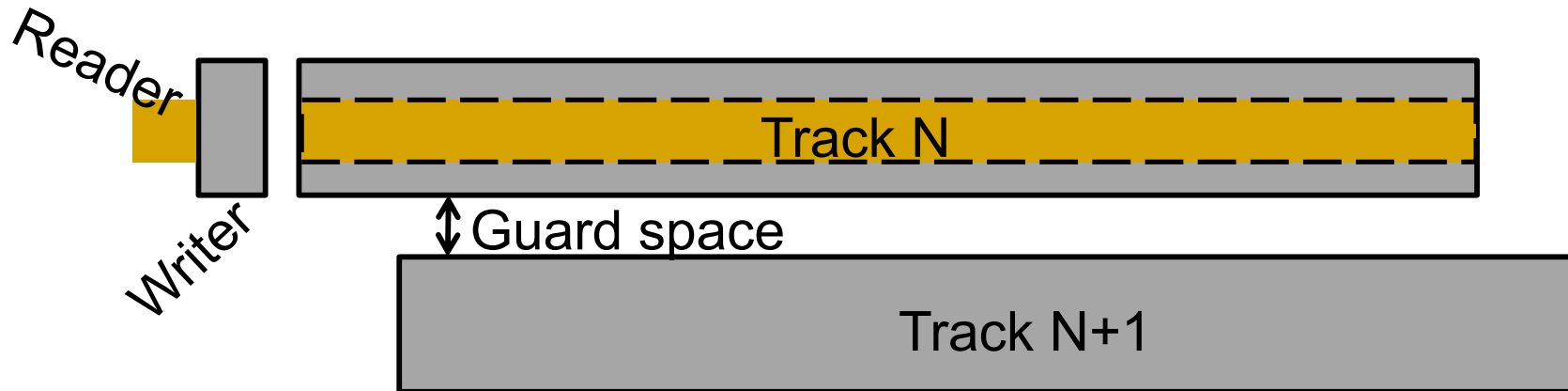
# Shingled Magnetic Recording (SMR)

- **Key:** Read head is **more precise** than write head.
- SMR is based on
  - **Writing** in a sequential way with tracks overlapped with the previous ones.
  - **Reading** the “exposed” data from shingled tracks.
- **Advantages** of Shingled Tracks:
  - Areal density↑, Capacity↑, and Cost↓
  - No major changes of disk are needed.
- **Design Challenge:**
  - Updating data to an existing track may **destroy** data on the subsequent tracks.
  - **SMR management** is needed.

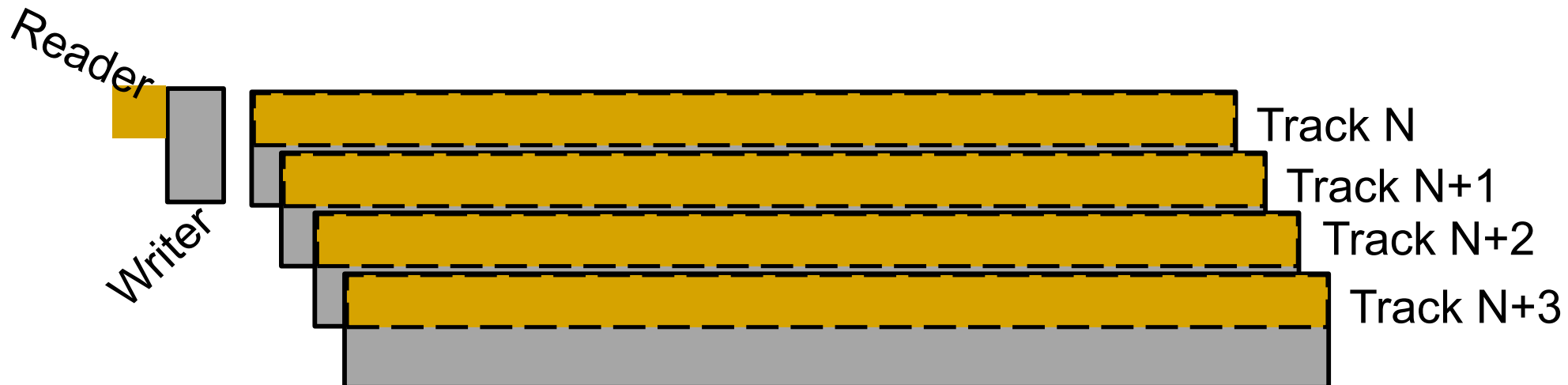


# Conventional vs. Shingled Writes

- Conventional Writes



- SMR Writes

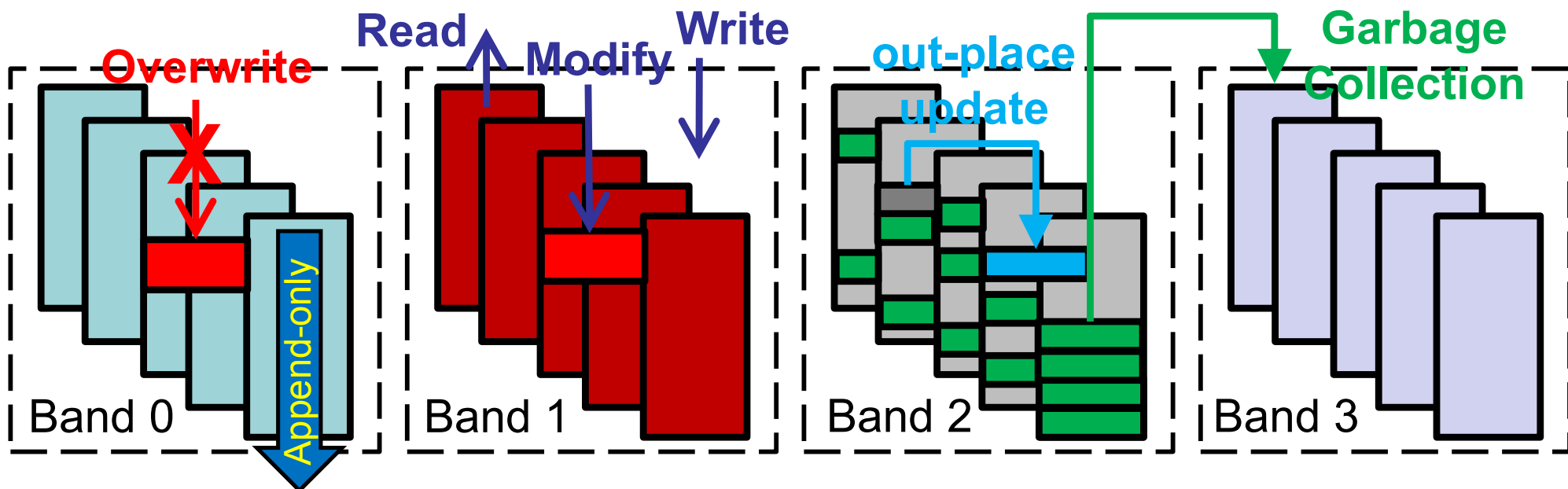


# SMR Inherent Challenges

- **Constraint:** Writes to SMR must be strictly **append-only** on the current write position (i.e., write pointer).
  - **Band-based management** can mitigate this challenge.
  - **Overwrite** is still prohibited in a band.

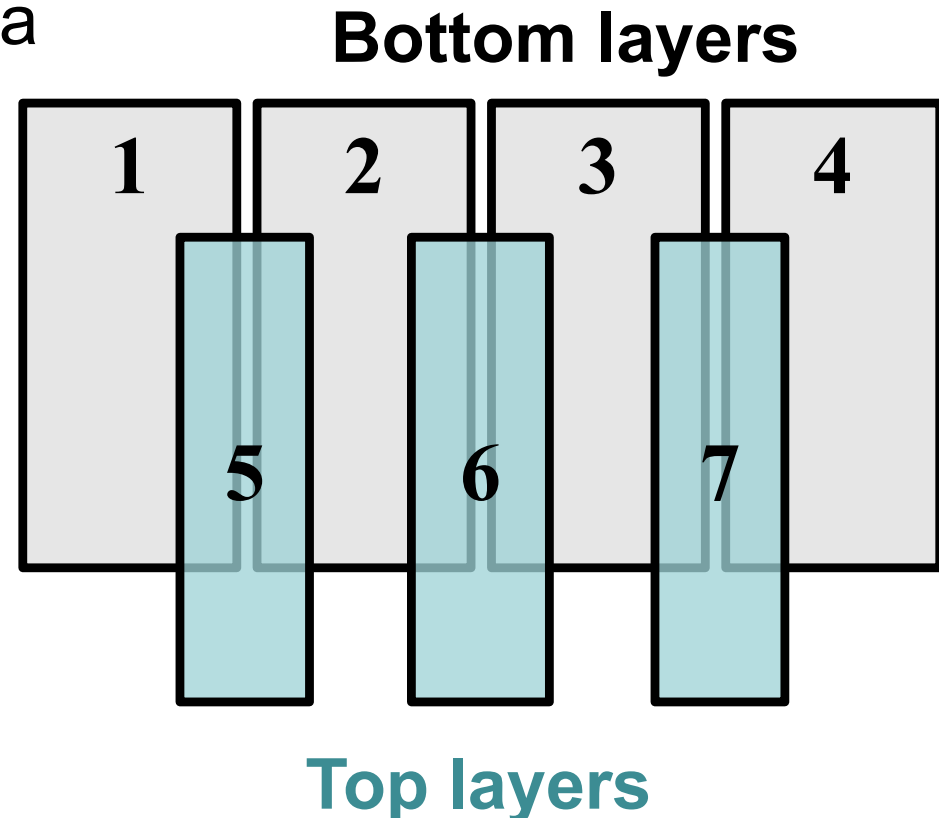
Approach 1) **Read-modify-write (RMW)** is expensive over a band.

Approach 2) Although **out-place-update** may service updates more efficiently, **address mapping** and **garbage collection** are needed.



# Interlaced Magnetic Recording

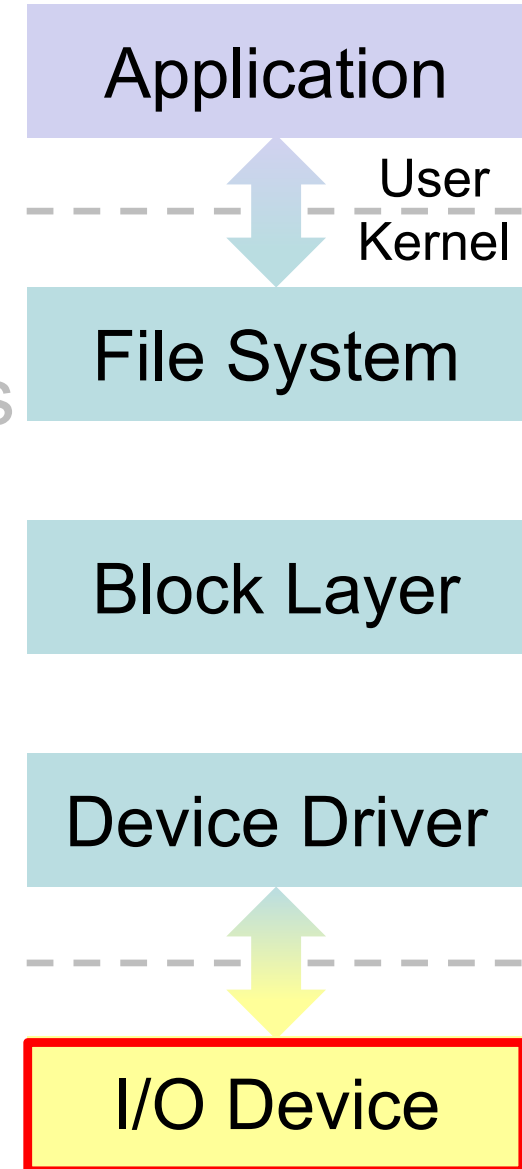
- Half of the tracks overlap
  - Bottom tracks are overlapped by top tracks
- Top tracks are narrower
  - Hold 80% - 90% as much data
- **No in-place updates are allowed for bottom tracks**
  - Solution: RMW or using a translation layer
    - Track-based Translation Layers for Interlaced Magnetic Recording [ATC '19]



# Agenda

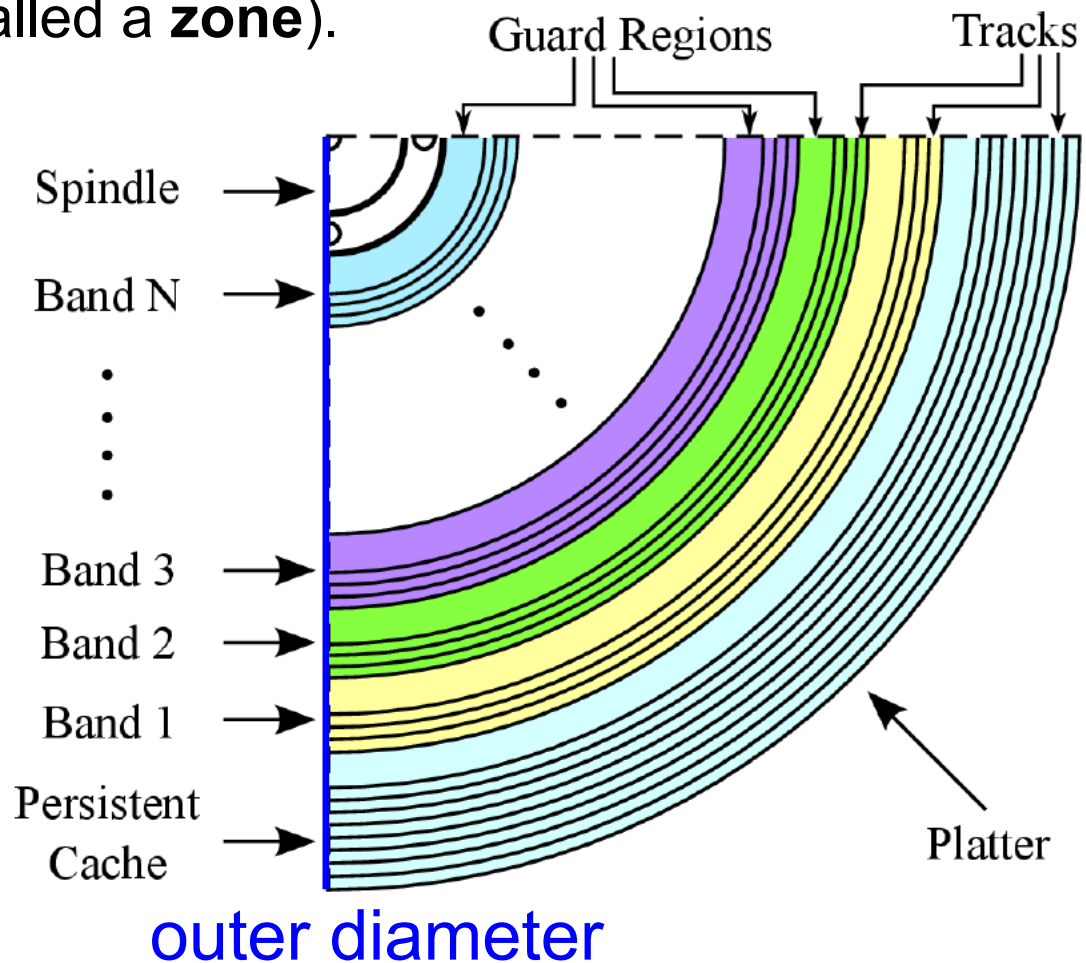
- Traditional Hard Disk Drive
  - Why and How
  - Development Bottleneck
- New Magnetic Recording Technologies
- Shingled Magnetic Recording (SMR)
  - Basics and Inherent Challenges
  - General Solution: Persistent Cache
  - Various SMR Drive Models and Designs
    - Drive-Managed SMR (DM-SMR)
    - Host-Aware SMR (HA-SMR)
    - Host-Managed SMR (HM-SMR)
    - Hybrid SMR

## I/O Stack



# General Solution to Non-Seq. Writes

- **Persistent Cache:** A small region to **stage non-sequential writes** to all bands (via out-place updates).
  - **Non-sequential Write:** Data are not written to the current “write pointer” of a **band** (also called a **zone**).
  - It can **postpone** updates and **reduce** the number of costly RMWs to bands.
  - It can be made by any **persistent storage** technology such as:
    - **Disk itself:** SMR tracks at **outer diameter (OD)**.
    - **Flash memory;** or
    - **Non-volatile memory.**



# Persistent Cache

- **Hybrid Mapping:**

- Persistent cache uses **extent mapping**;
- Disk space is mapped at **band granularity**.

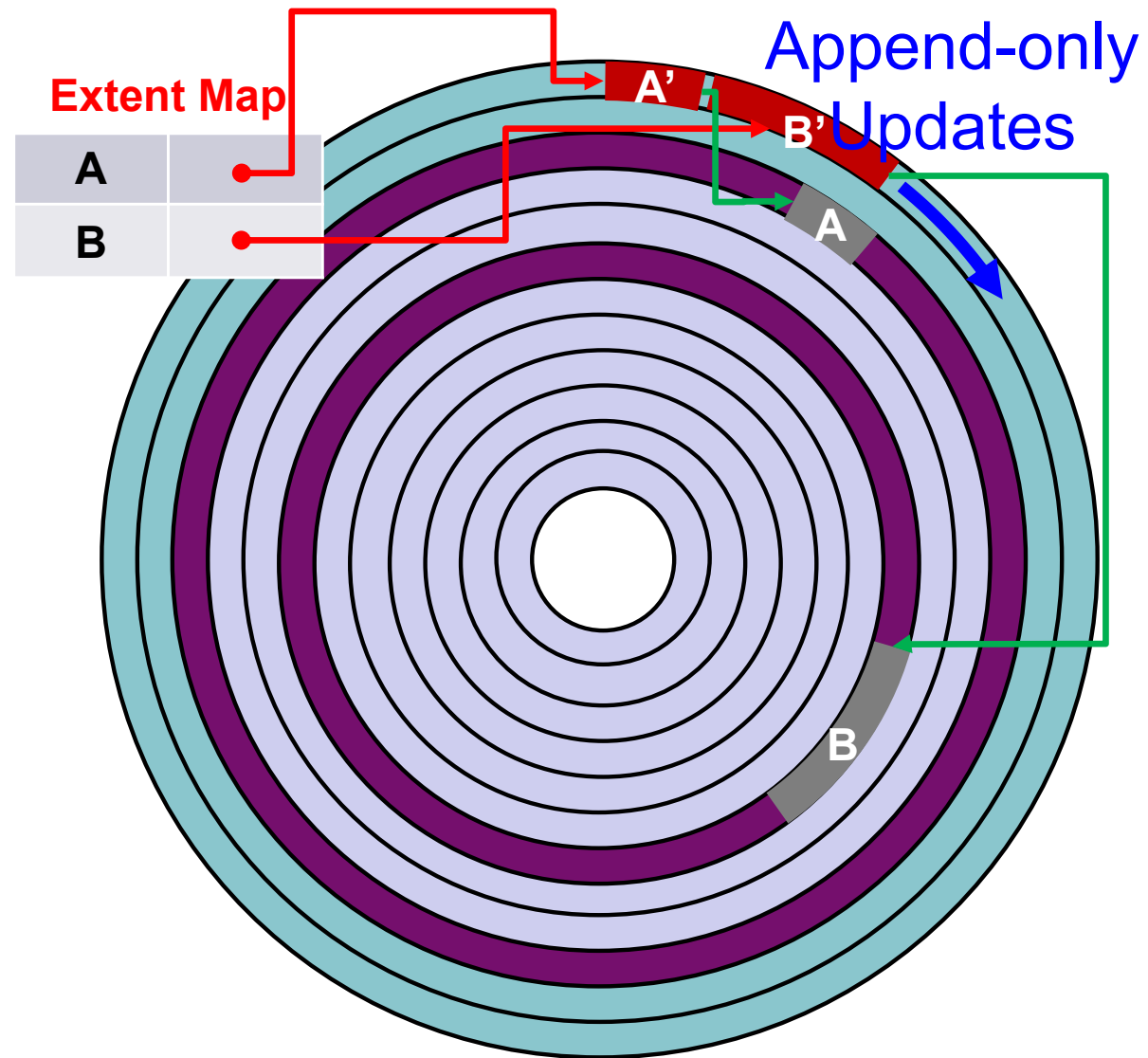
- **Garbage Collection** is needed to read-modify-write a band.

- **Lazy Cleaning**

- Start until the persistent cache is **almost full**

- **Aggressive Cleaning**

- Start as soon as disk **idleness** is detected



Band are shown in **purple**;  
Persistent cache is shown in **green**.

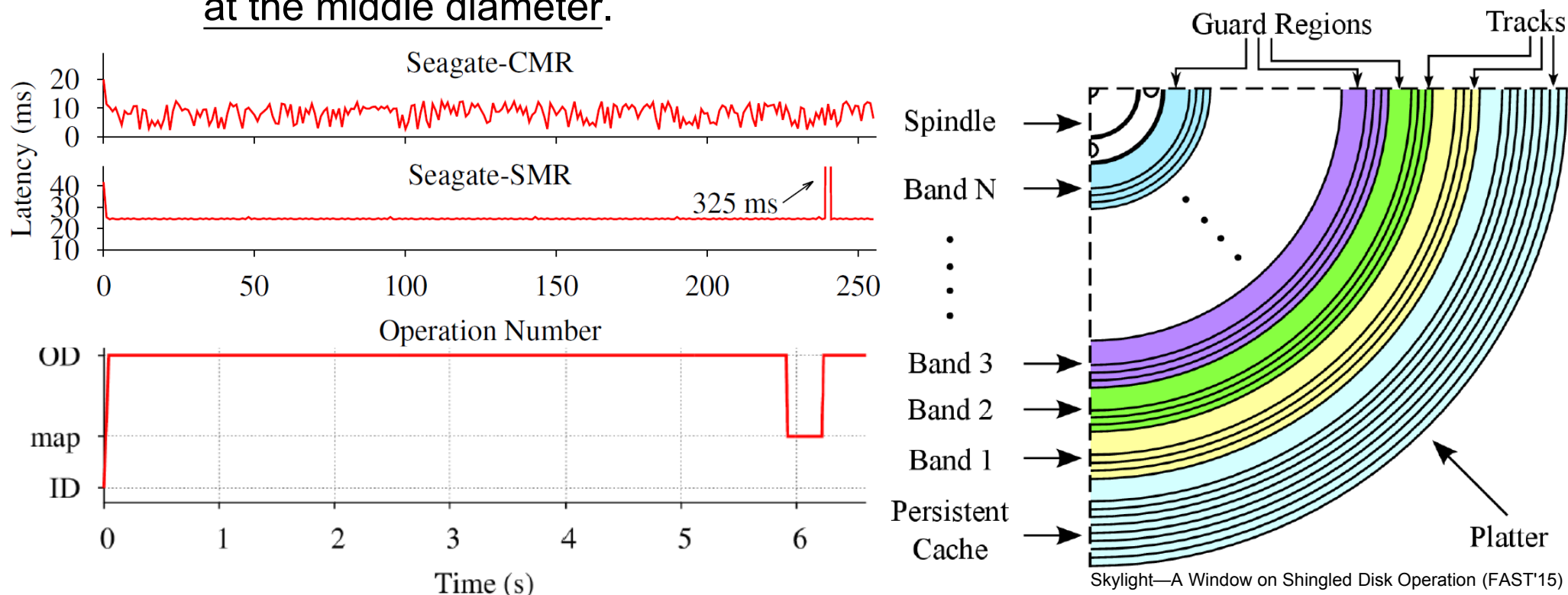
# “Observational” Results (1/4)

- **Idea:** Reverse engineer key properties of a DM-SMR drive.
  - **Software:** Launch crafted I/O operations using fio.
  - **Hardware:** Install a “skylight” on the drive to track the head movements using a high-speed camera.
- Real DM-SMR drive was tested.
  - Seagate ST5000AS0011
  - 5900RPM (rotation time 10 ms)
  - Four platters
  - Eight heads
  - 5TB capacity



# “Observational” Results (2/4)

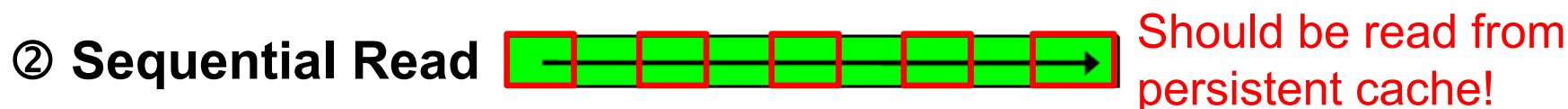
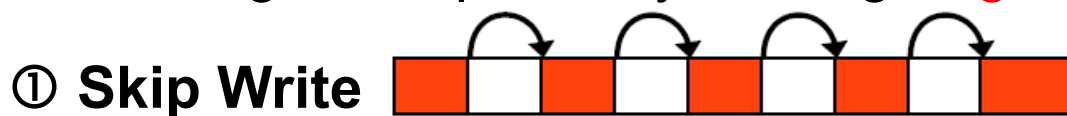
- Random Write Test: Write the first GB blocks randomly.
- **Observational Results**:
  - Seagate-CMR shows **varying latencies** (due to repositions).
  - Seagate-SMR shows a **fixed  $\approx 25\text{ms}$  latency** with a **bump**.
    - **Fixed Latency**: There is a **persistent cache** at the outer diameter.
    - **Bump at the 240th Write**: There is (likely) a **persistent cache map** stored at the middle diameter.



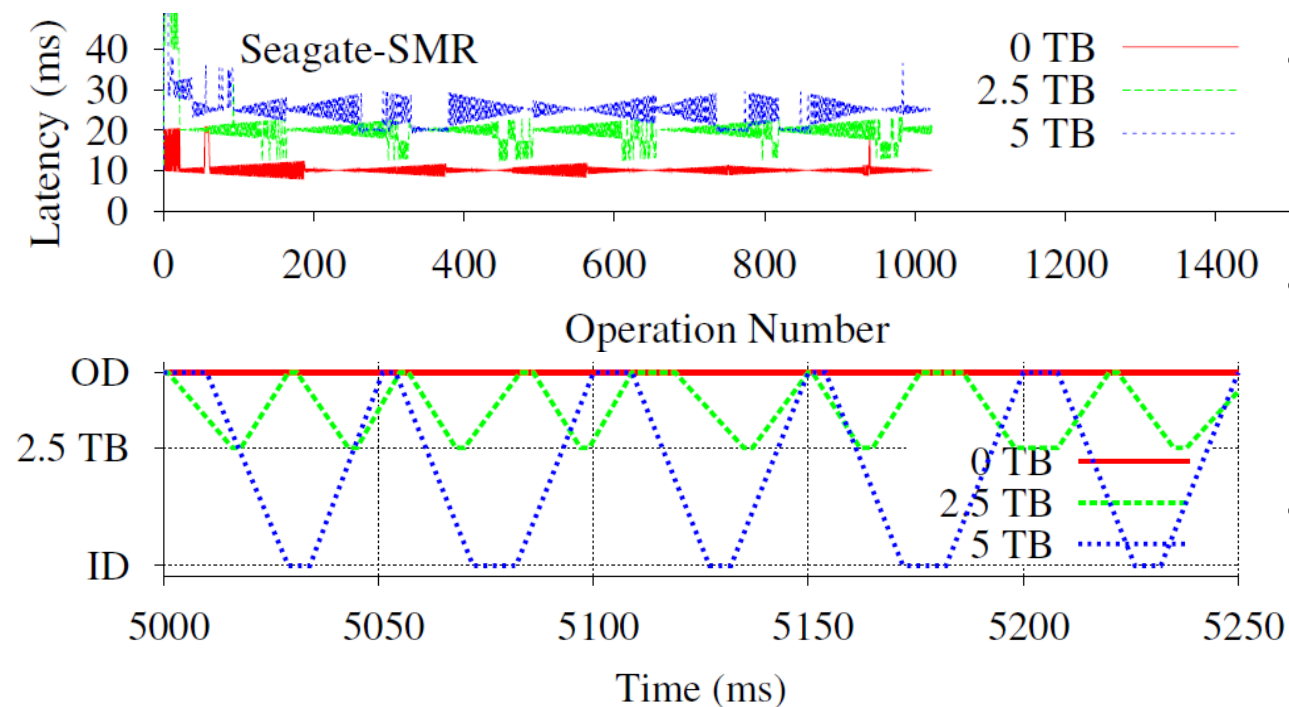
# “Observational” Results (3/4)

- Fragmented Read Test:

- ① Choose a small region and writing every other block in it
- ② Read the region sequentially, forcing **fragmented reads**



- **Observational Results:**



- A fragmented read at **low LBAs** (0 TB) incurs **negligible seek**;
- A fragmented read at **high LBAs** (5 TB) incurs **high seek**.
- The persistent cache locates **at the OD**.

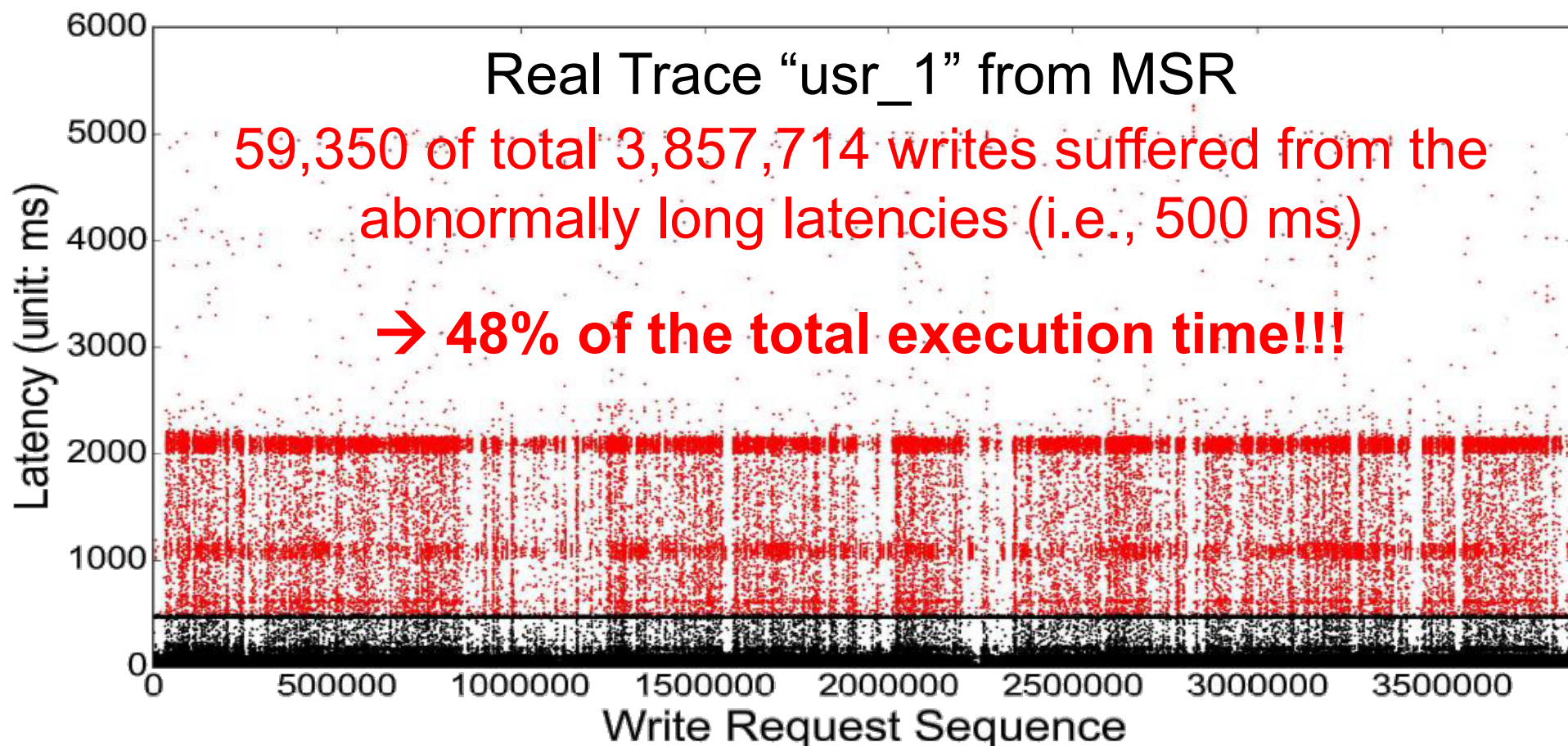
# “Observational” Results (4/4)

- All properties discovered by Skylight methodology:

| Property                  | Drive Model       |                   |
|---------------------------|-------------------|-------------------|
|                           | ST5000AS0011      | ST8000AS0011      |
| Drive Type                | SMR               | SMR               |
| Persistent Cache Type     | Disk              | Disk              |
| Cache Layout and Location | Single, at the OD | Single, at the OD |
| Cache Size                | 20 GiB            | 25 GiB            |
| Cache Map Size            | 200,000           | 250,000           |
| Band Size                 | 17–36 MiB         | 15–40 MiB         |
| Block Mapping             | Static            | Static            |
| Cleaning Type             | Aggressive        | Aggressive        |
| Cleaning Algorithm        | FIFO              | FIFO              |
| Cleaning Time             | 0.6–1.6 s/band    | 0.6–1.6 s/band    |
| Zone Structure            | 4–20 GiB          | 5–40 GiB          |
| Shingling Direction       | Towards MD        | N/A               |

# “Evaluated” Results

- Long latency issue is observed under real workloads.
  - Real workloads were replayed on a Seagate 8TB HA-SMR.
  - Persistent cache cleaning may be the bottleneck.

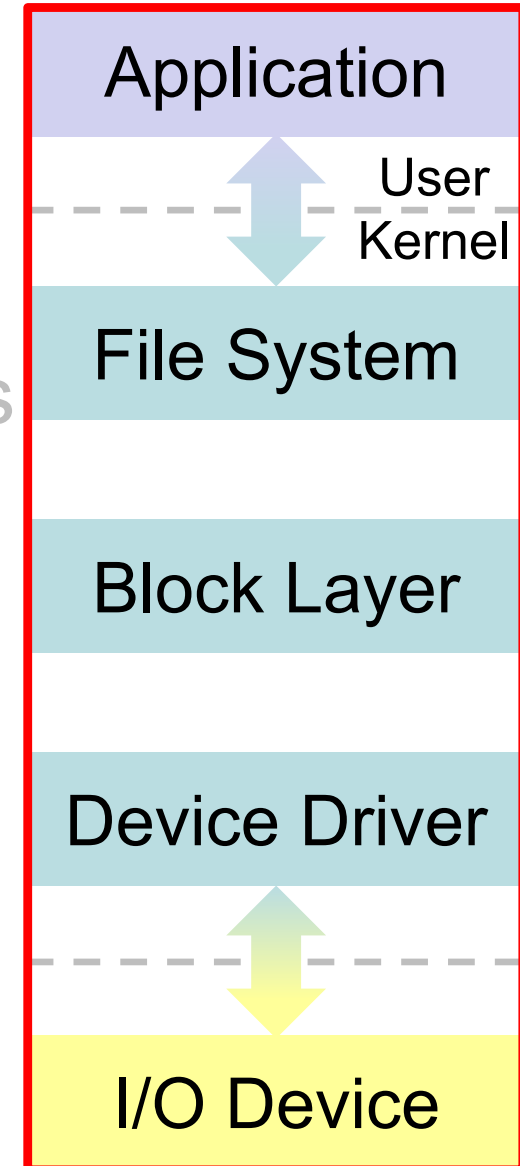


**Persistent cache is not the panacea!**

# Agenda

- Traditional Hard Disk Drive
  - Why and How
  - Development Bottleneck
- New Magnetic Recording Technologies
- Shingled Magnetic Recording (SMR)
  - Basics and Inherent Challenges
  - General Solution: Persistent Cache
  - Various SMR Drive Models and Designs
    - Drive-Managed SMR (DM-SMR)
    - Host-Aware SMR (HA-SMR)
    - Host-Managed SMR (HM-SMR)
    - Hybrid SMR

## I/O Stack



# Who Should Take Care of SMR



Technical Committee T10

- **T10** defines three possible models:

- **Host-Managed (HM)**

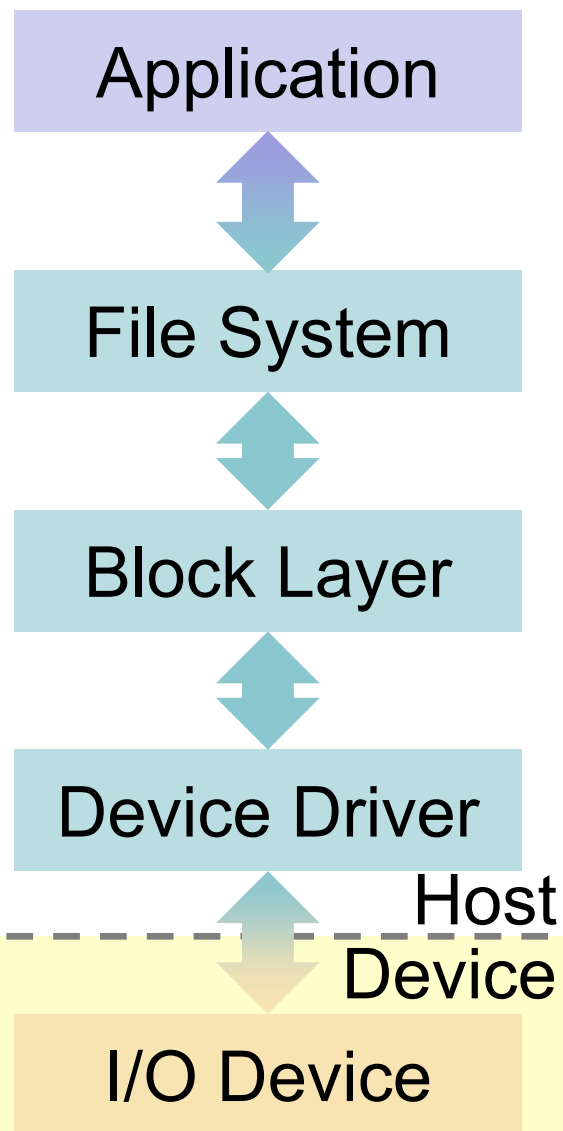
- Host must write a zone **sequentially**
  - “Non-sequential write” is **prohibited**
- Require lots of **system software redesigns**
- + **High predictability** on raw I/O performance

- **Host-Aware (HA)**

- + Host is suggested to write a zone sequentially
  - “Non-sequential write” is **handled** by drive
- Require moderate **system software redesigns**

- **Drive-Managed (DM)**

- + **Transparent** to the host side
  - **Drop-in replacement** for traditional drives
  - A **firmware** handles “non-sequential write”
- **Low predictability** on I/O performance of drives



# Various SMR Drive Models

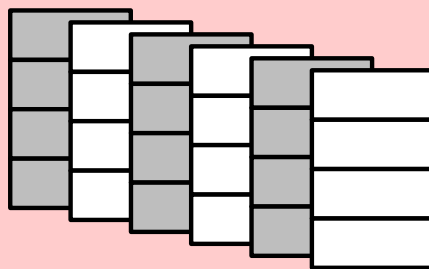


Technical Committee T10



**Non-sequential  
Writes**

**Persistent  
Cache**

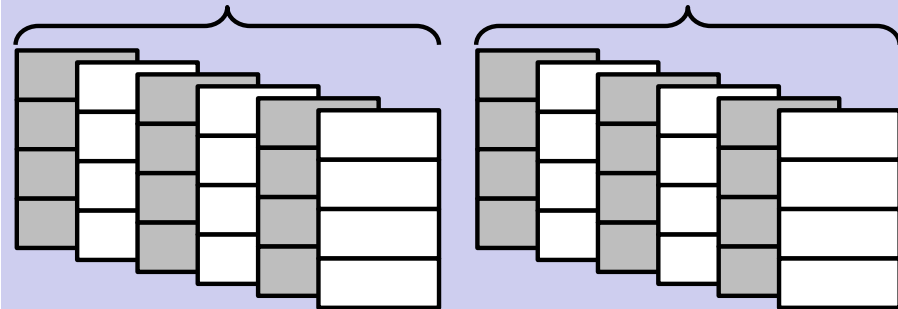


**Sequential  
Writes**

**SMR Zones**

Zone 0

Zone 1



**SMR Drive Model**

**Persistent Cache**

**SMR Zone**

**Host-Managed (HM)**

**X**

**O**



**Host-Aware (HA)**

**Δ**  
*(not accessible by  
the host)*

**O**

**Drive-Managed (DM)**

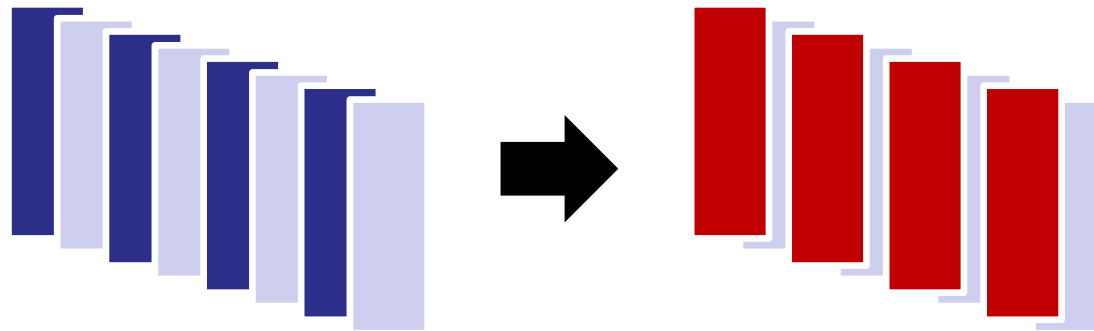
**Δ**  
*(transparent to host)*

**Δ**  
*(transparent to host)*



# Case Study of DM-SMR: SMaRT

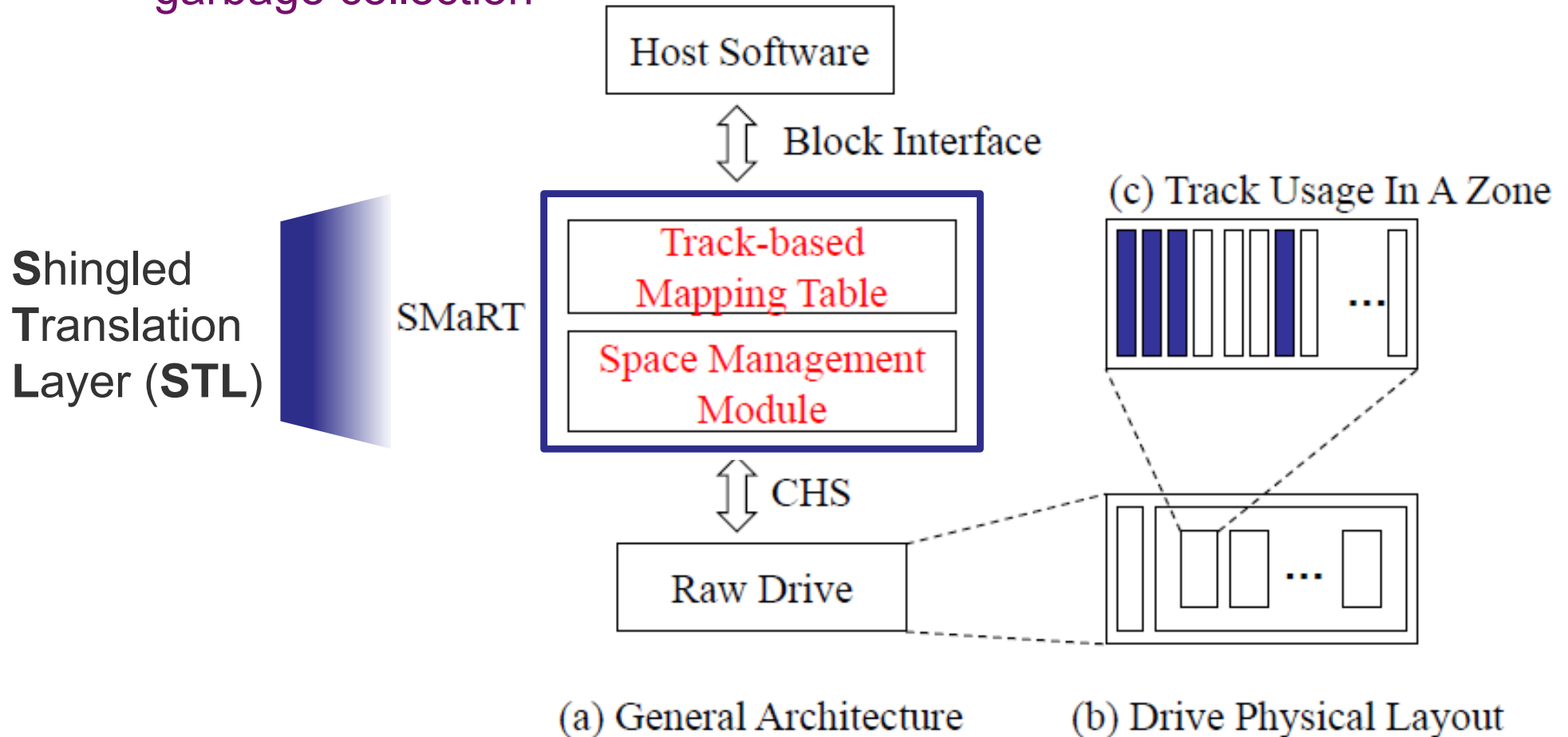
- DM-SMR can manage all details of the drive to:
  - Improve overall I/O performance
  - **Remove or mitigate the use of persistent cache**
- **Observation:** A track supports in-place update if its following track is free or contains stale data.



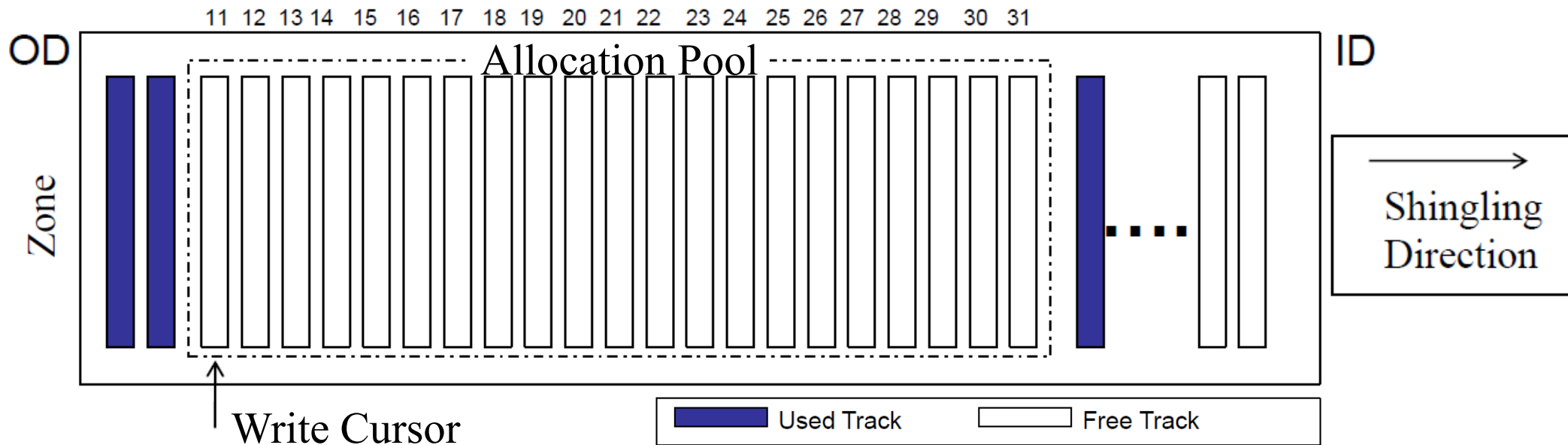
- **Track-based Management:**
  - **Sector** is too small: Sector-based mapping creates **huge mapping table** and introduces **garbage blocks/sectors**.
  - **Track** is moderate: Tracks can be **managed more flexibly**.
  - **Zone** is too big: Non-sequential writes are **troublesome**.

# SMaRT: Overall Architecture

- Two major modules in the **firmware** (i.e., **STL**):
  - A **track-based mapping** to support **track-level translation**
  - A **space management module** to manage free **track allocations** and **garbage collection**



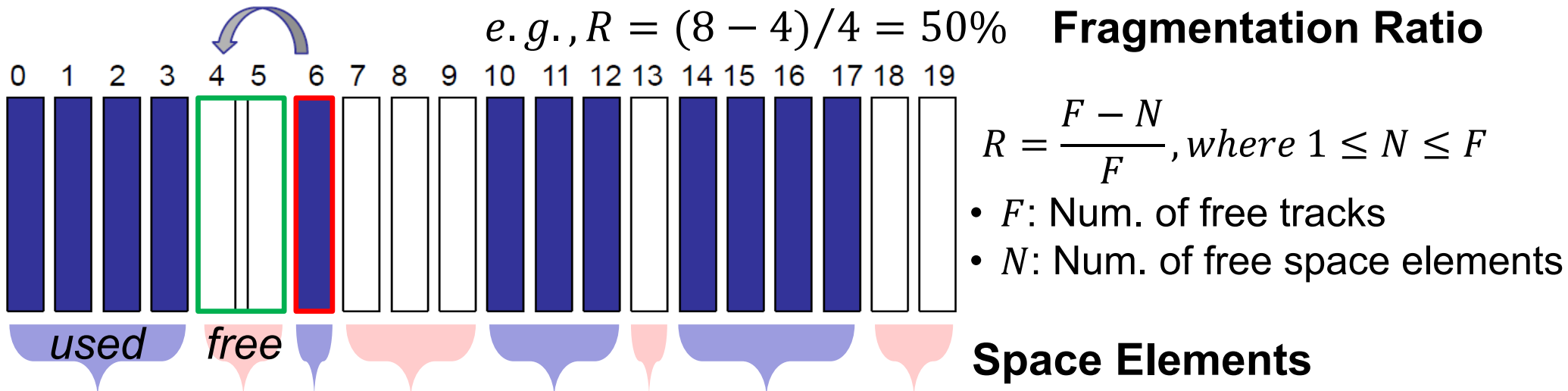
# SMaRT: Track Allocation in a Zone



- All writes (new data and updated data) go to the **write cursor** of an **allocation pool** sequentially.
- Newly updated tracks are deemed as **hot data**.
  - SMaRT allocates an extra track as **safety gap** for each hot track if space utilization is less than 50%.
- When the current pool is full, choose the new one of the **largest** number of **consecutive free tracks**.

# SMaRT: Garbage Collection

- SMaRT invokes GCs in each zone whenever the **fragmentation ratio  $R$**  is smaller than a *threshold*:



## ① Pick Victim

- Search for a **victim space element** in a zone (from the leftmost to rightmost), where its size is smaller than the *used-to-free ratio  $W$*

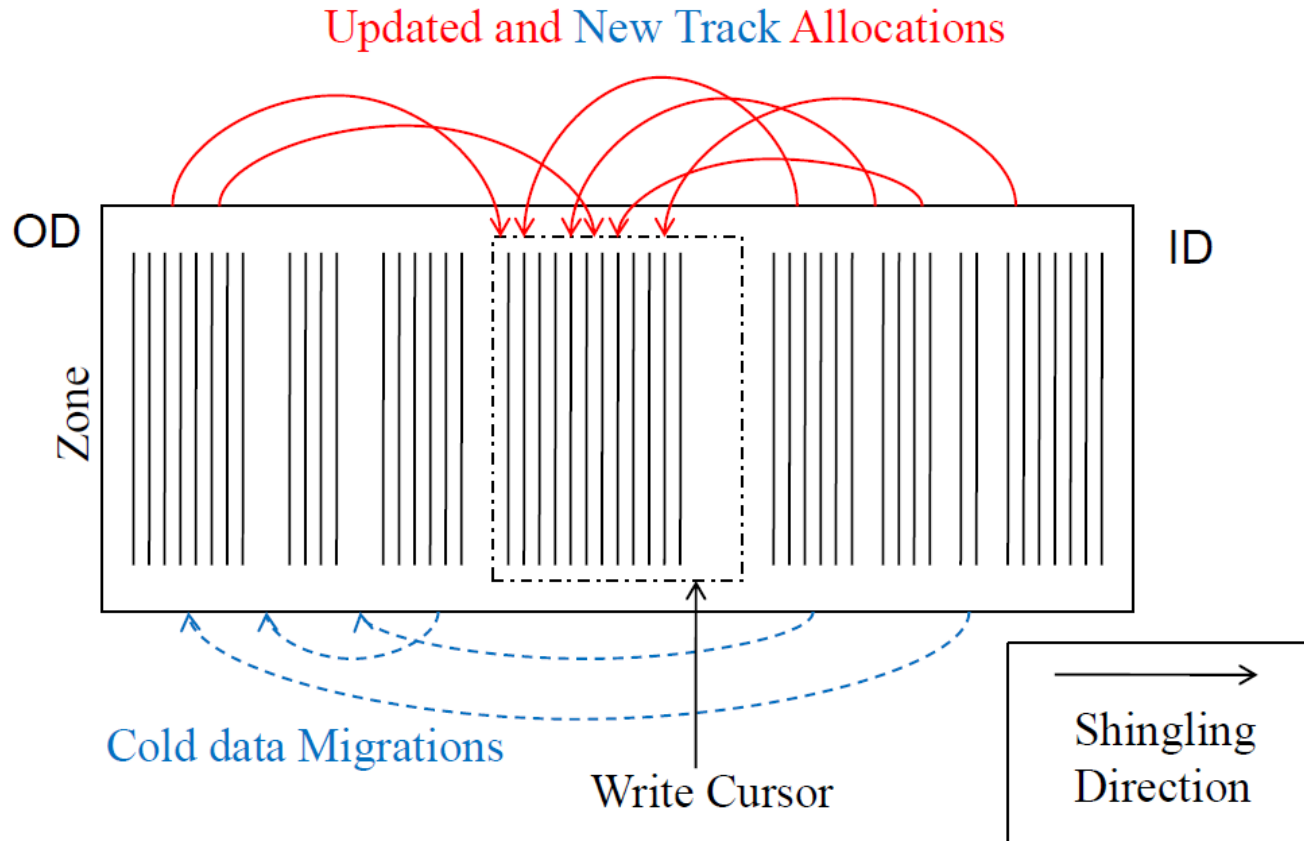
$$W = \frac{U}{1 - U}, \text{ where } U \text{ is the drive space usage}$$

## ② Pick Destination

- Allocate to the **first free space element to the left** that fits it; or
- **Shift left** and append to its left neighbor

# SMaRT: Auto Cold Data Progression

- The free track allocation of GC provides a good opportunity of **automatic cold data progression**.
  - The **cold** data will mostly **stay at the left side**, while **hot** data gets updated and pushed to **the right side** of the zones



- Unnecessary cold data movements can be **avoided**.

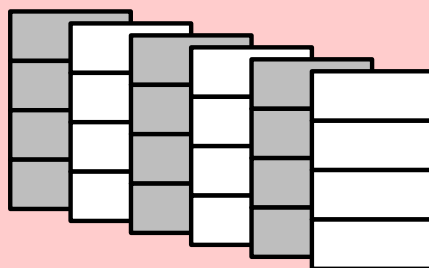
# Various SMR Drive Models

**incits**  
Technical Committee T10



**Non-sequential  
Writes**

**Persistent  
Cache**

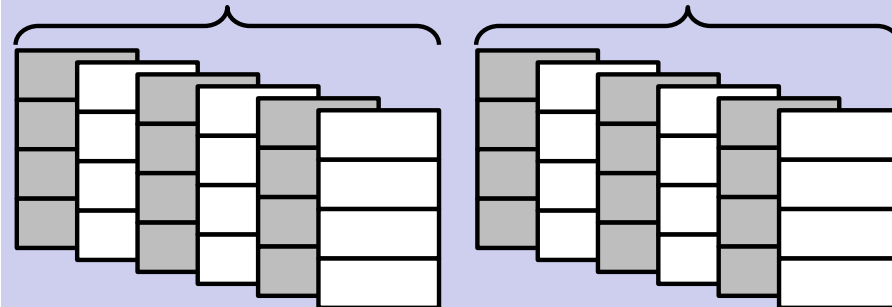


**Sequential  
Writes**

**SMR Zones**

Zone 0

Zone 1



**SMR Drive Model**

**Persistent Cache**

**SMR Zone**

**Host-Managed (HM)**

**X**

**O**



**Host-Aware (HA)**

**Δ**  
*(not accessible by  
the host)*

**O**

**Drive-Managed (DM)**

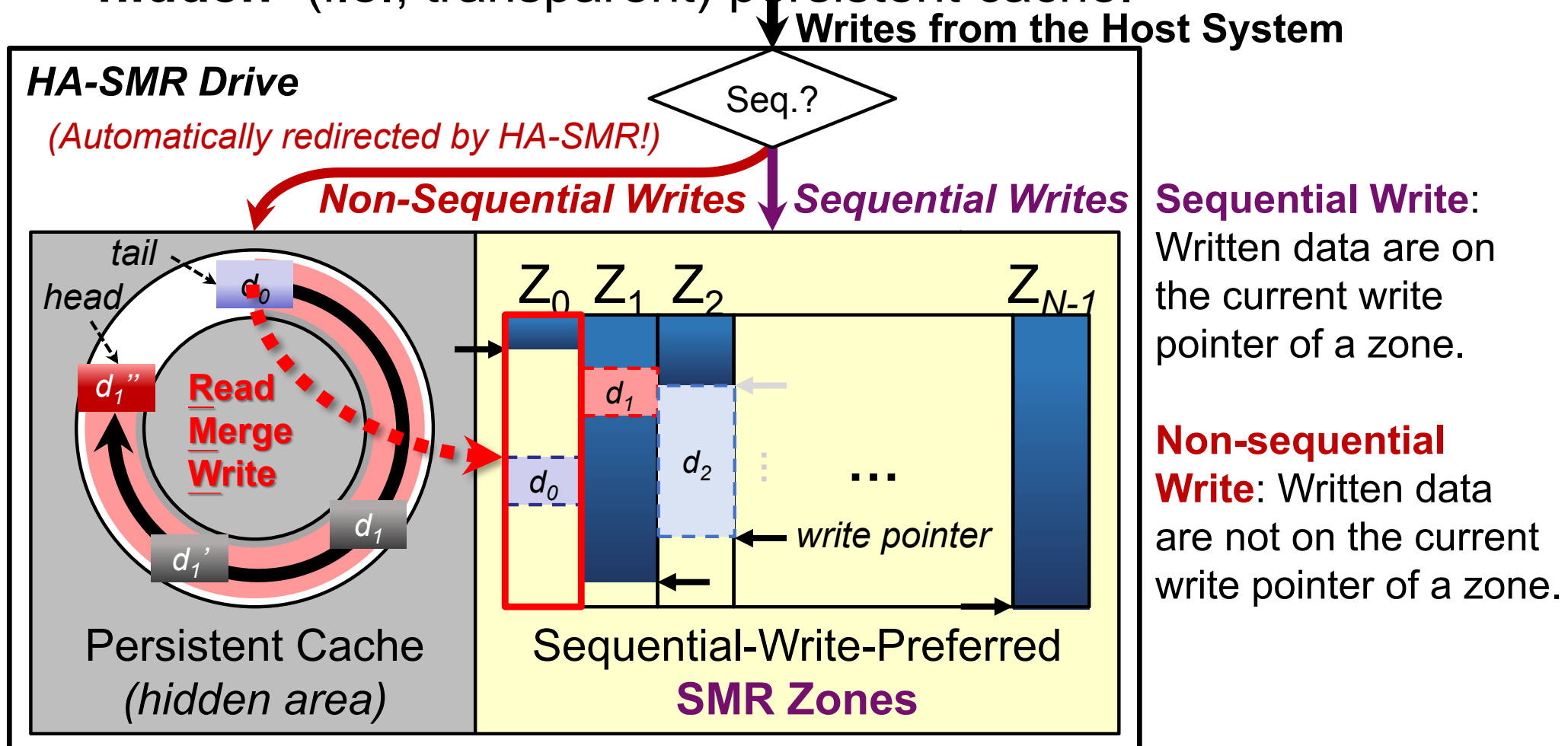
**Δ**  
*(transparent to host)*

**Δ**  
*(transparent to host)*



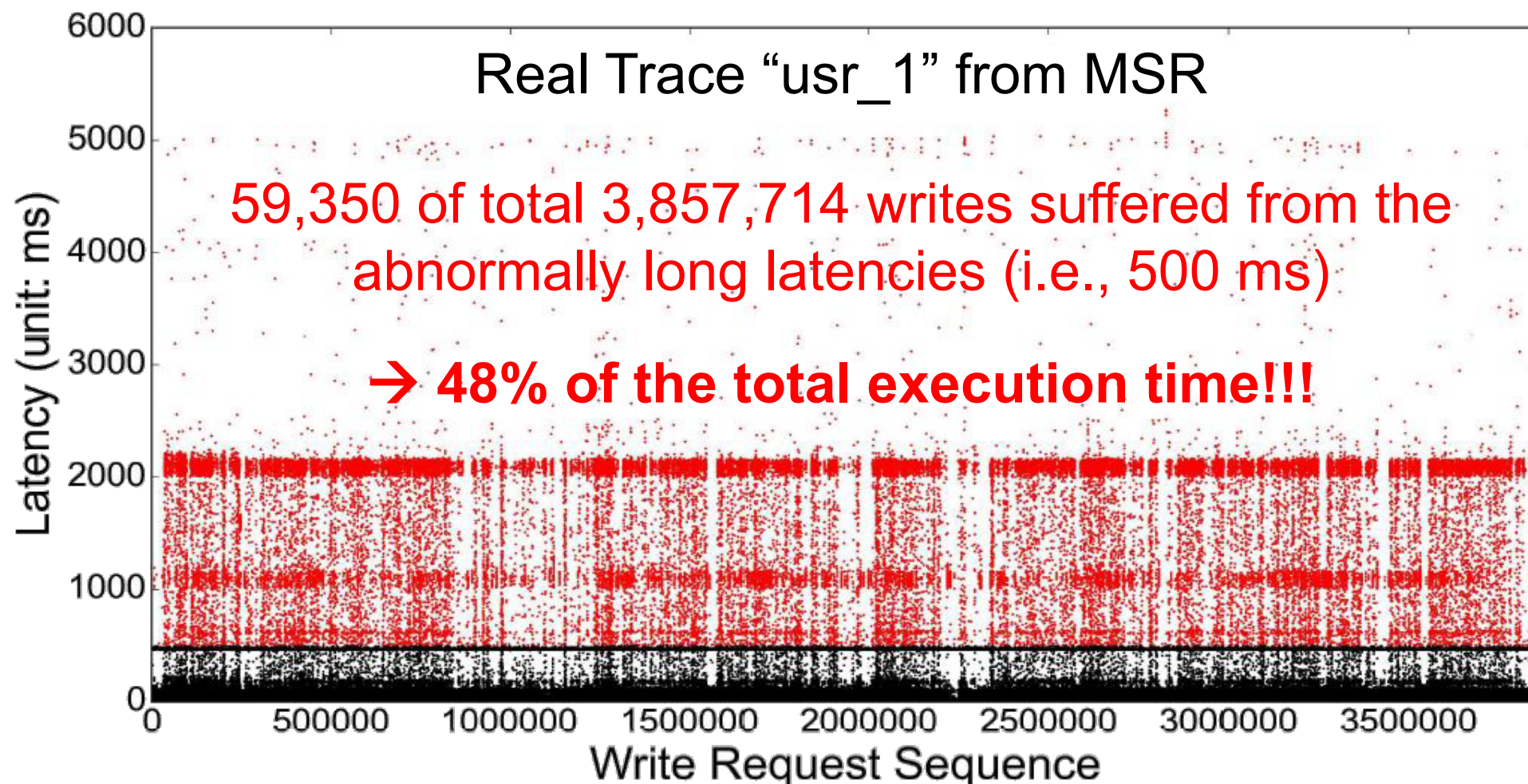
# Case Study of HA-SMR: VPC

- HA-SMR drive reports “**zone information**” to the host,
- HA-SMR automatically handles **non-sequential writes** in a “**hidden**” (i.e., transparent) persistent cache.



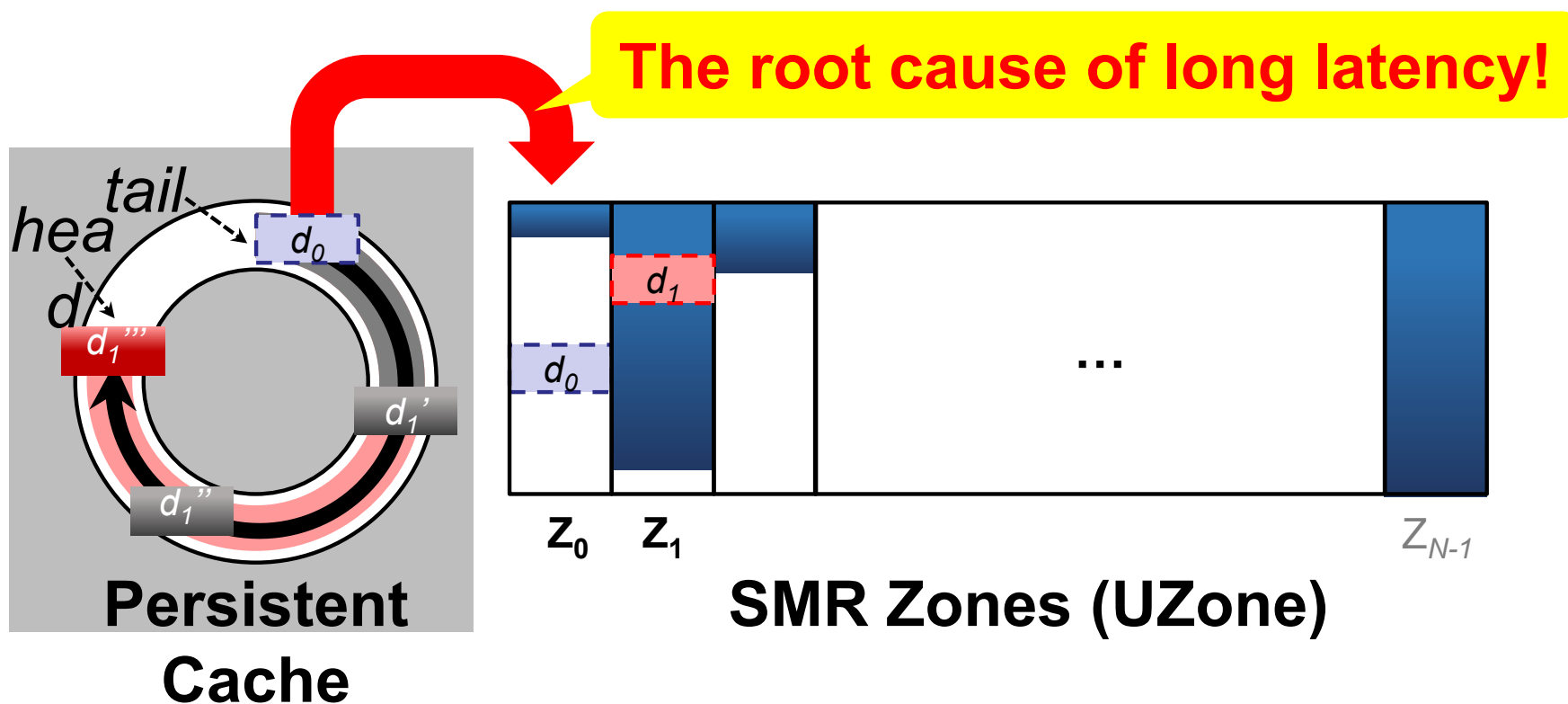
# Long Latency Behavior of HA-SMR

- **Long latency issue** is observed under real workloads.
  - Real workloads were replayed on a Seagate 8TB HA-SMR.
  - Persistent cache cleaning may be the **bottleneck**.



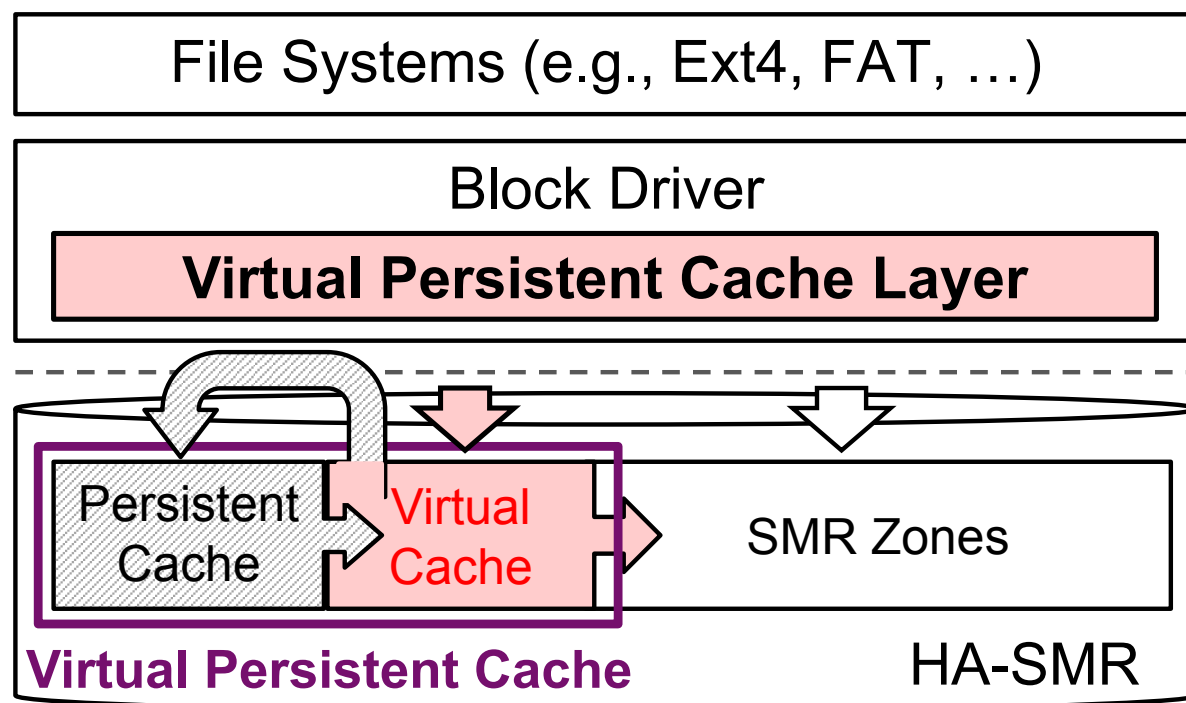
# Root of Long Latency Behavior

- **Fact:** **Some** are updated more frequently than **others**.
  - Non-sequential writes of **different update frequencies** are **mixed** in the persistent cache.
  - Read-merge-writes must **often** merge back **non-frequently-updated data** residing at the tail of the persistent cache.



# Virtual Persistent Cache for HA-SMR

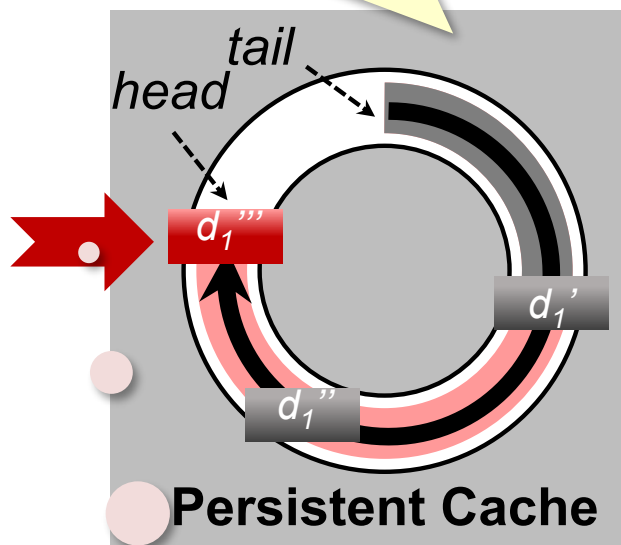
- **VPC:** **Virtually** enlarge the **persistent cache**
  - Virtual Persistent Cache = Persistent Cache + **Virtual Cache**
  - **Goal:** **Avoid overwhelming** the persistent cache by non-sequential writes of a wide range of update frequencies.
- Virtual Cache
  - Take a few zones based on the needs.
    - Trade little space for great performance!
  - Take advantage of the **computing and memory resources** of the host system.



*In practice, the proposed design can be realized at the block drive layer as a general solution for various applications.*

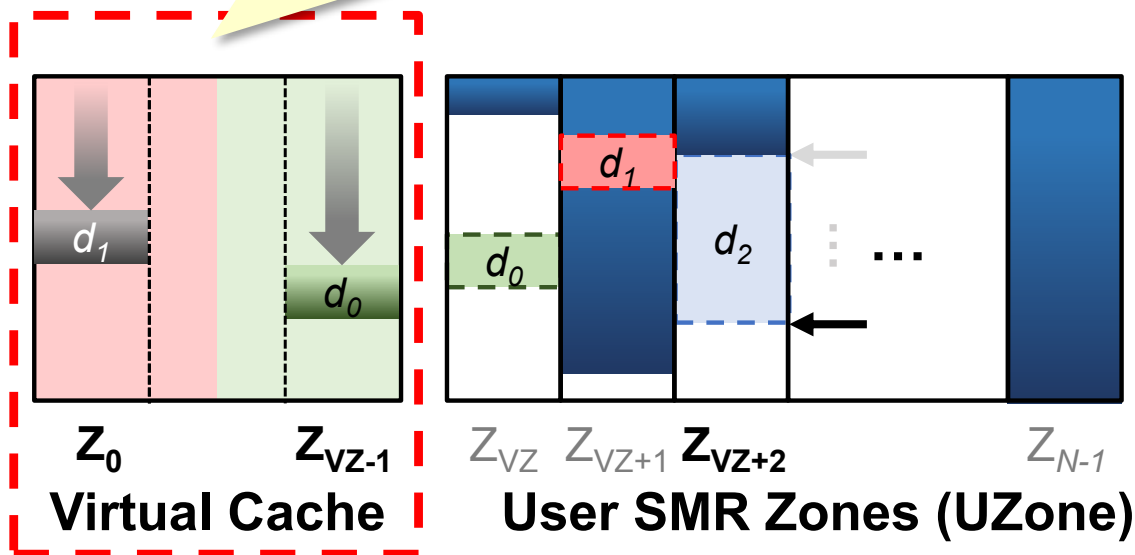
# VPC: Host-Drive Collaboration

(3) Only allow **freq. updated data** to be redirected into Persistent Cache



Drive

(2) Always **sequentially accommodate** the received non-seq. writes into Virtual Cache by **remapping** the LBA



Host

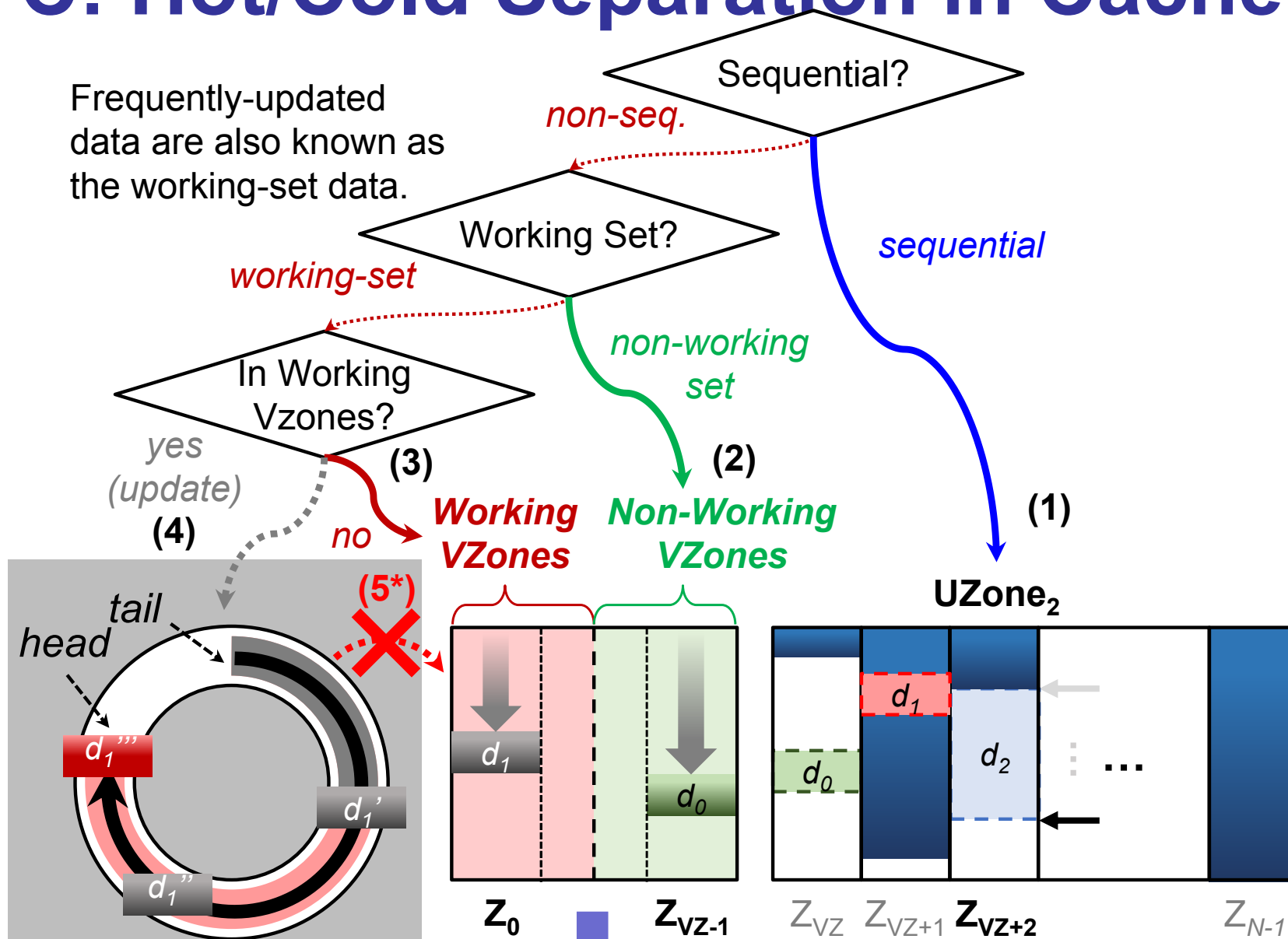
Host

Making it a **non-seq. update** to  $d_1$  in Virtual Cache!

(1) **Adaptively allocate** few SMR zones to work as Virtual Cache

# VPC: Hot/Cold Separation in Cache(s)

Frequently-updated data are also known as the working-set data.

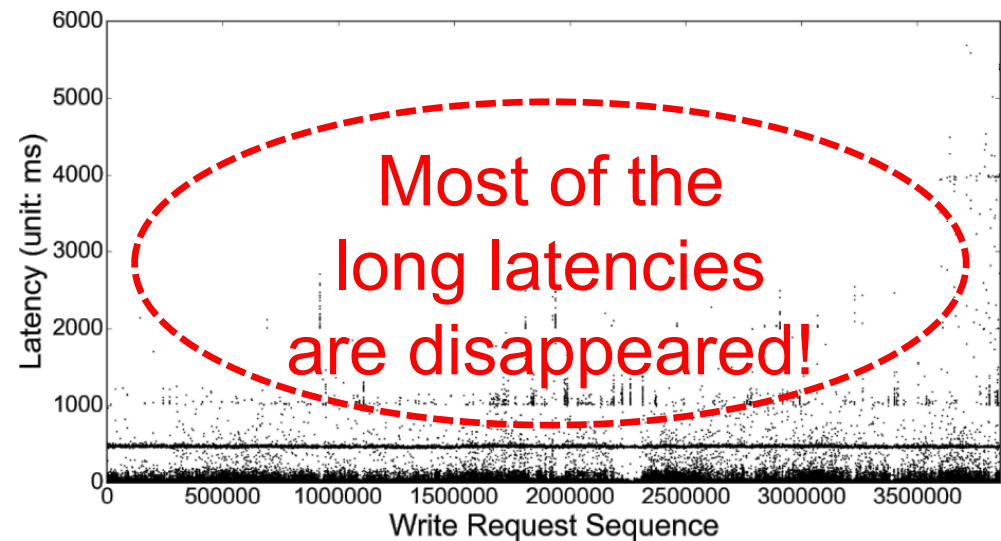
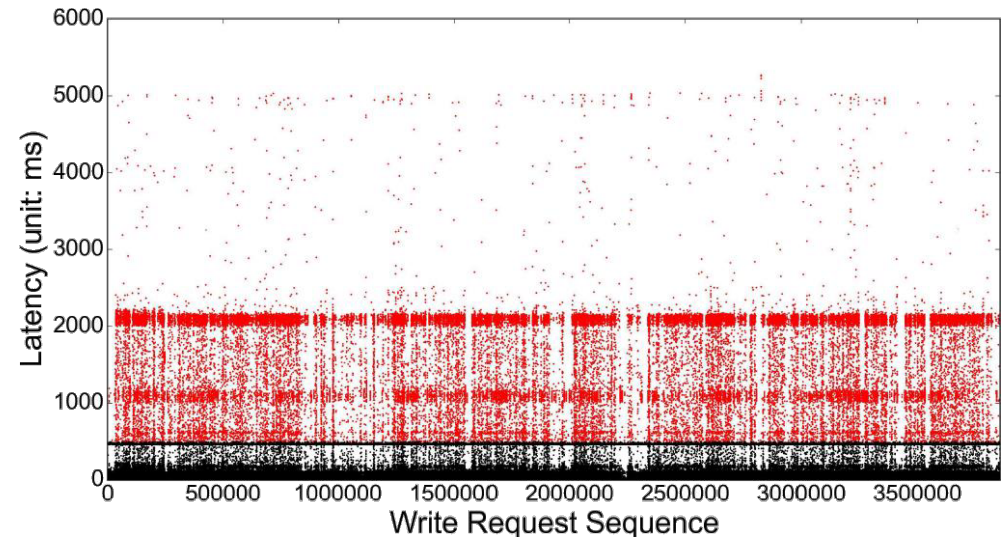
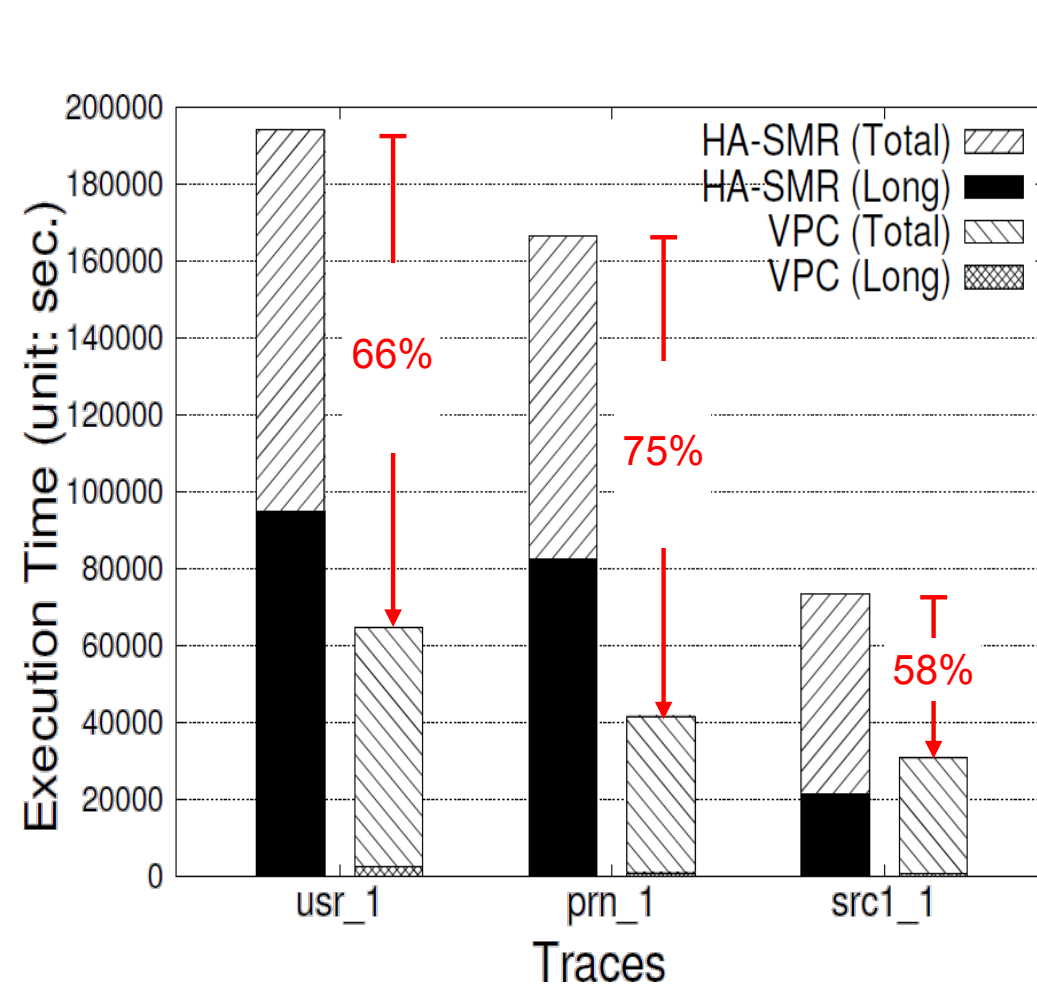


**Avoiding overloading persistent cache to eliminate most RMWs.**

**The host can clean virtual cache with limited RMWs.**

# VPC: Evaluation Results

- Great Improvement of Total Execution Time



# Various SMR Drive Models

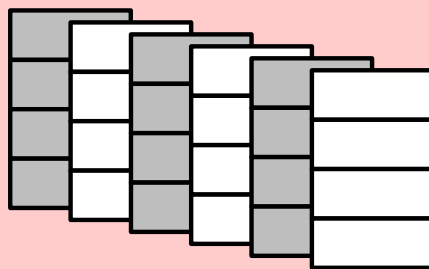


Technical Committee T10



**Non-sequential  
Writes**

**Persistent  
Cache**

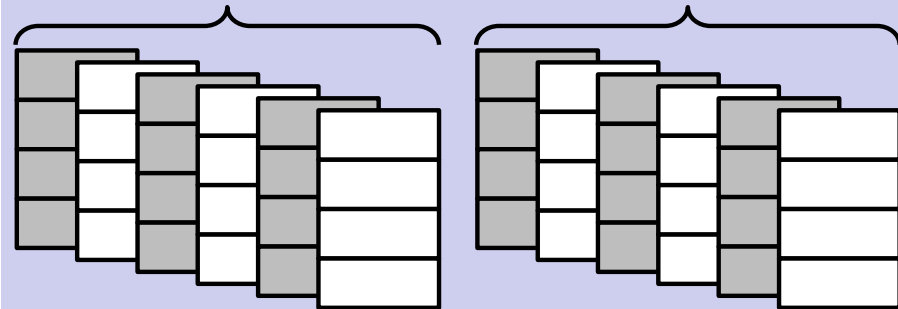


**Sequential  
Writes**

**SMR Zones**

Zone 0

Zone 1



**SMR Drive Model**

**Persistent Cache**

**SMR Zone**

**Host-Managed (HM)**

**X**

**O**



**Host-Aware (HA)**

**Δ**  
*(not accessible by  
the host)*

**O**

**Drive-Managed (DM)**

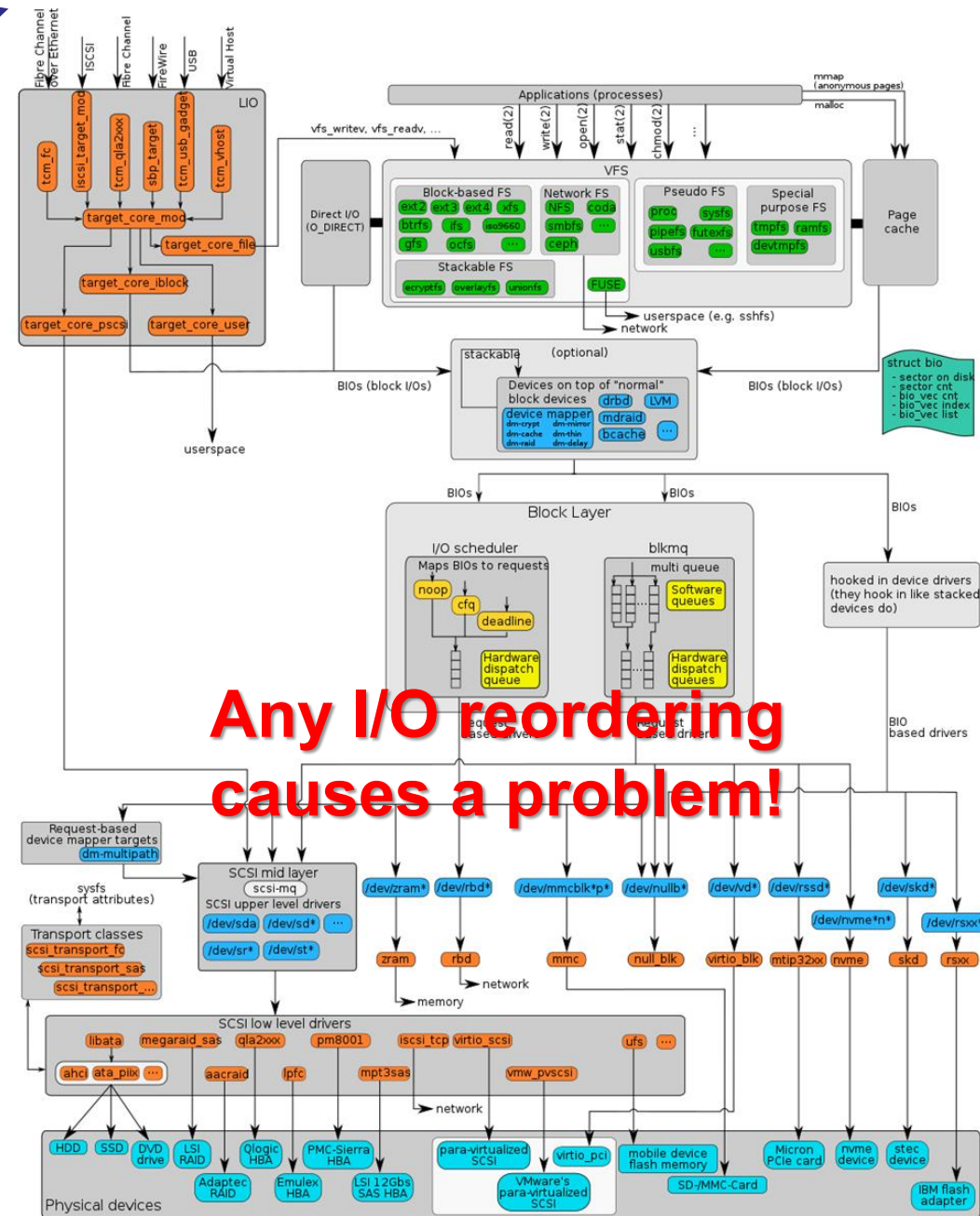
**Δ**  
*(transparent to host)*

**Δ**  
*(transparent to host)*



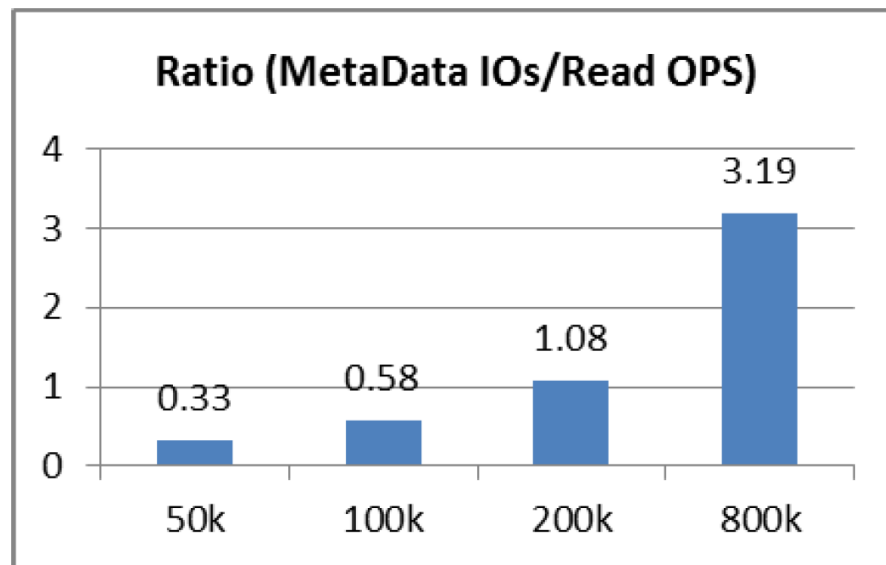
# Challenges of HM-SMR

- The host **must** write a zone sequentially.
  - Non-sequential write is **strictly prohibited**.
- **All of layers** must be **SMR compatible**.
  - A **dedicated FS** must be designed for SMR.
- The host also knows more about the system.
  - Possible to substantially improve the **performance** and **I/O predictability**.



# Case Study of HM-SMR: HiSMRfs

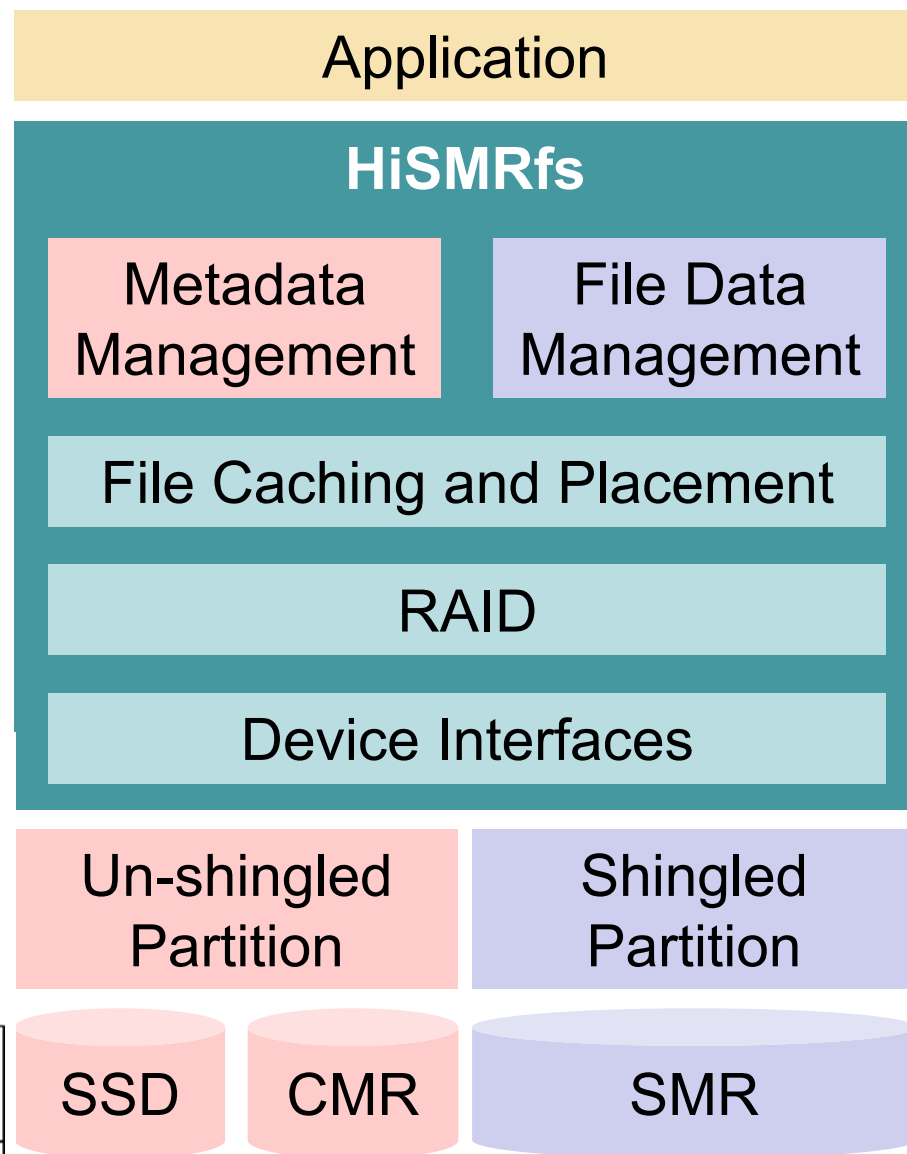
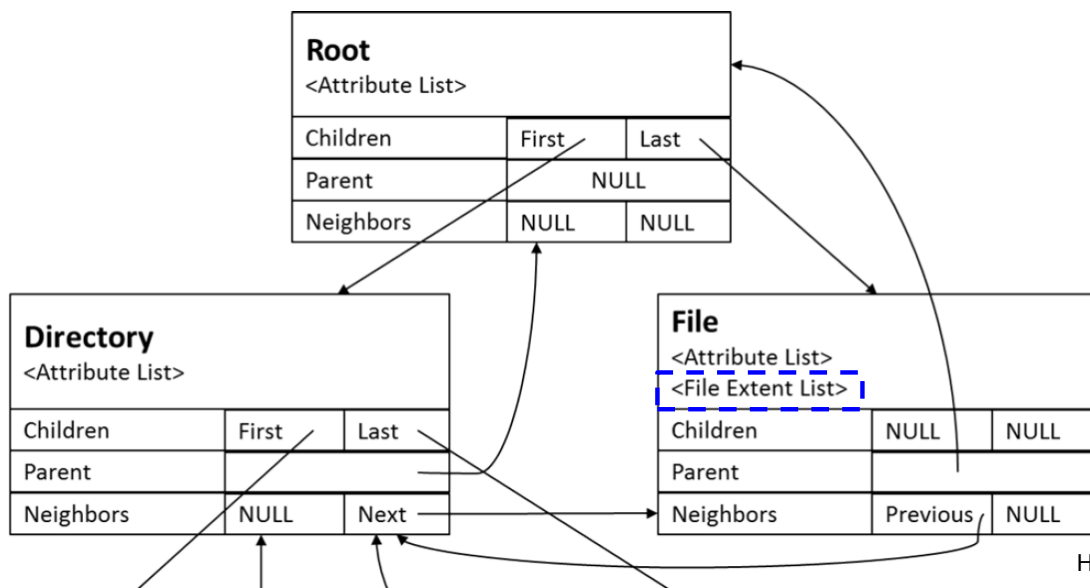
- **HiSMRfs** is **high performance FS** for SMR drives.
  - It can manage **SMR zones** and **support random writes** without remapping layer implemented inside SMR drives.
    - Random Writes? Always writing at the end through appending.
  - It further **separates data and metadata** storage, and manages them differently to achieve high performance.
    - **Observation:** As the number of files increases, a **larger number of metadata IOs** is needed to access a single file data.



With 800K files stored, reading a data file requires, on average, **three** metadata accesses.

# HiSMRfs: Overall Architecture

- The **metadata** and **data** are managed **separately**.
  - **Meta: Un-shingled Partitions**
    - **File extent list** indicates where the file contents are kept in SMR.
  - **File Data: Shingled Partitions**
    - *File Caching and Placement module further **caches** hot file data in un-shingled partitions.*



# HiSMRfs: File Data Management

- HiSMRfs is a SMR-compliant file system.
  - File data are **sequentially appended** at the end of **each SMR zone** (which is used as a **data log**).
  - File Data Management has **four** major modules:
    - ① The **File Data Allocation** module determines **where the data will be written**.
    - ② The **File Request Queuing and Scheduling** module arranges file **read/write requests into queues**.
    - ③ The **Garbage Collection** module is responsible to **reclaim released space from zones**.
      - File deletion and modification will cause invalid data blocks in the data log, and these invalid blocks need to be reclaimed.
    - ④ The **Zone Layout** module **emulates a SMR data layout** and its related information.

# Wrap-up: Who Should Take Care of SMR



Technical Committee T10

- **T10** defines three possible models:

- **Host-Managed (HM):** e.g., HiSMRfs

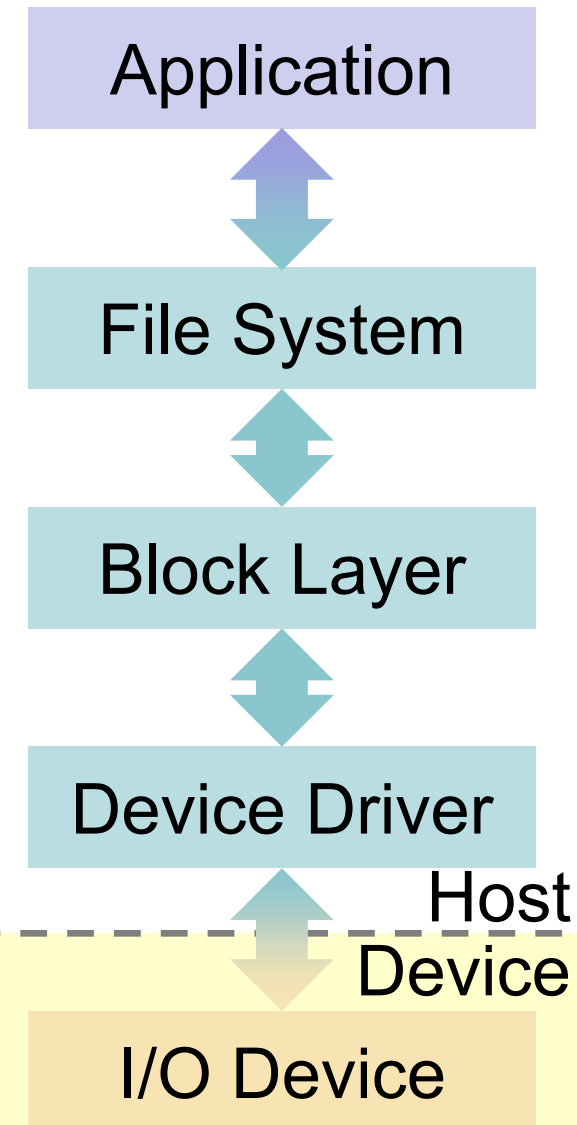
- Host must write a zone **sequentially**
  - “Non-sequential write” is **prohibited**
- Require lots of **system software redesigns**
- + **High predictability** on raw I/O performance

- **Host-Aware (HA):** e.g., VPC

- + Host is suggested to write a zone sequentially
  - “Non-sequential write” is **handled** by drive
- Require moderate **system software redesigns**

- **Drive-Managed (DM):** e.g., SMaRT

- + **Transparent** to the host side
  - **Drop-in replacement** for traditional drives
  - A **firmware** handles “non-sequential write”
- **Low predictability** on I/O performance of drives

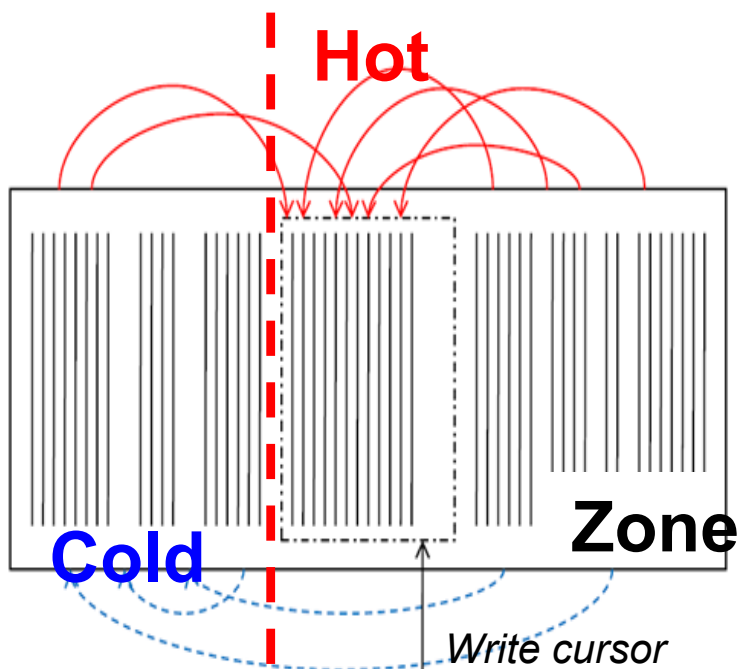


# Wrap-up: Hot/Cold Separation

- These designs share one key technique, **hot/cold separation**, to mitigate the **non-sequential write** issue.
- They achieve it by different policies and **granularities**:

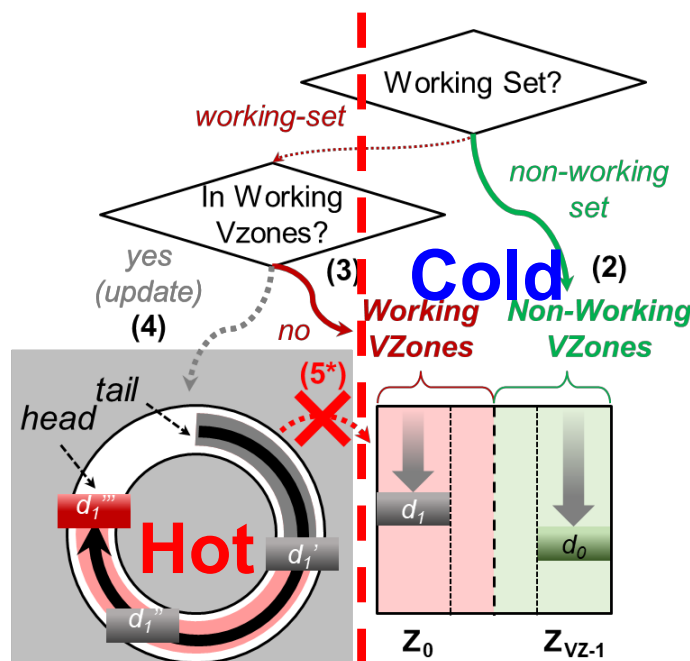
## ➤ HM-SMR: SMaRT

- **Hot Data**: RHS tracks of a **zone**
- **Cold Data**: LHS tracks of a **zone**



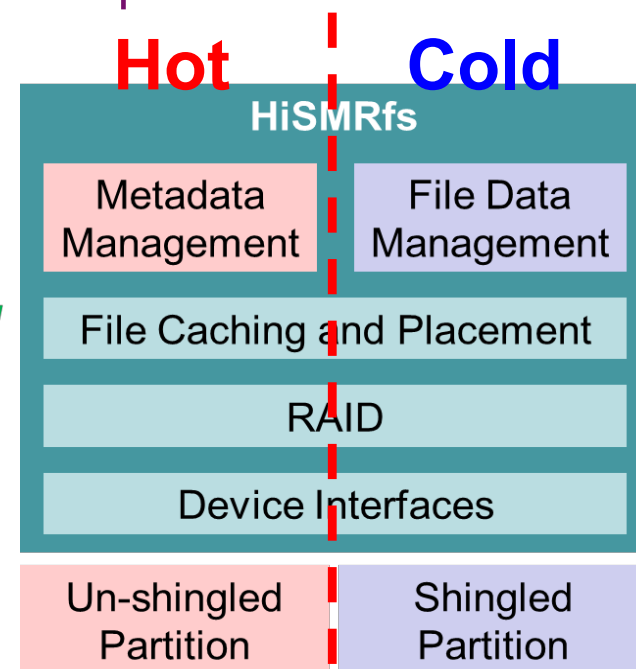
## ➤ HA-SMR: VPC

- **Hot Data**: Persistent cache
- **Cold Data**: Virtual cache



## ➤ HM-SMR: HiSMRfs

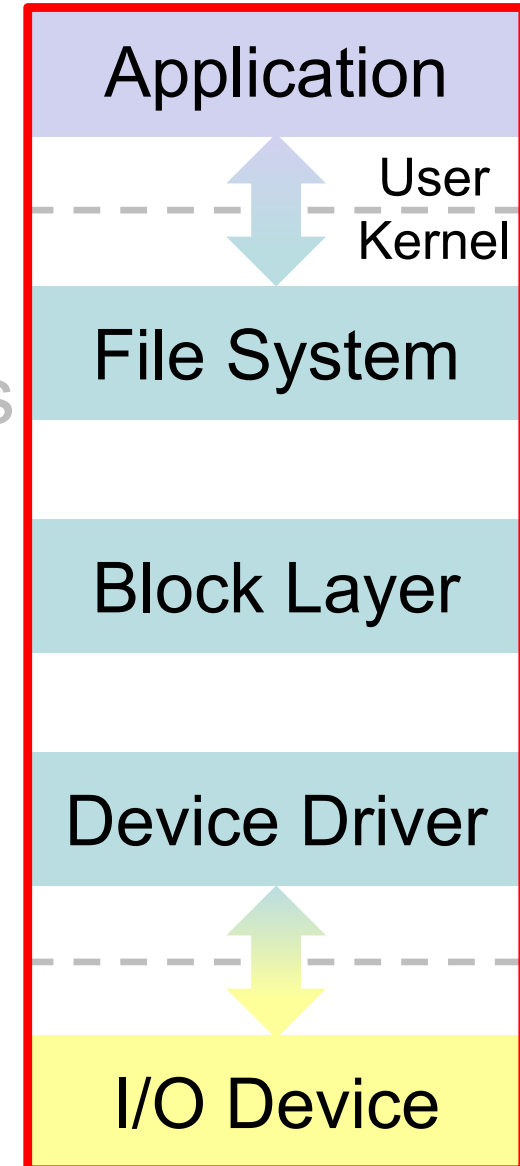
- **Metadata**: Un-shingled **partition**
- **Data**: Shingled **partition**



# Agenda

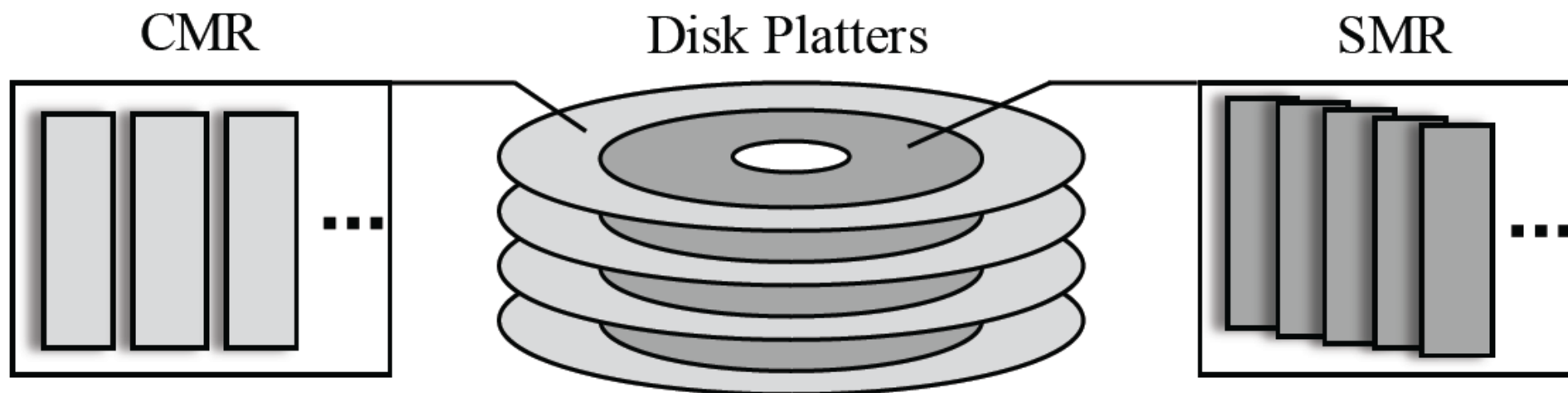
- Traditional Hard Disk Drive
  - Why and How
  - Development Bottleneck
- New Magnetic Recording Technologies
- Shingled Magnetic Recording (SMR)
  - Basics and Inherent Challenges
  - General Solution: Persistent Cache
  - Various SMR Drive Models and Designs
    - Drive-Managed SMR (DM-SMR)
    - Host-Aware SMR (HA-SMR)
    - Host-Managed SMR (HM-SMR)
    - Hybrid SMR

## I/O Stack



# Hybrid SMR (H-SMR)

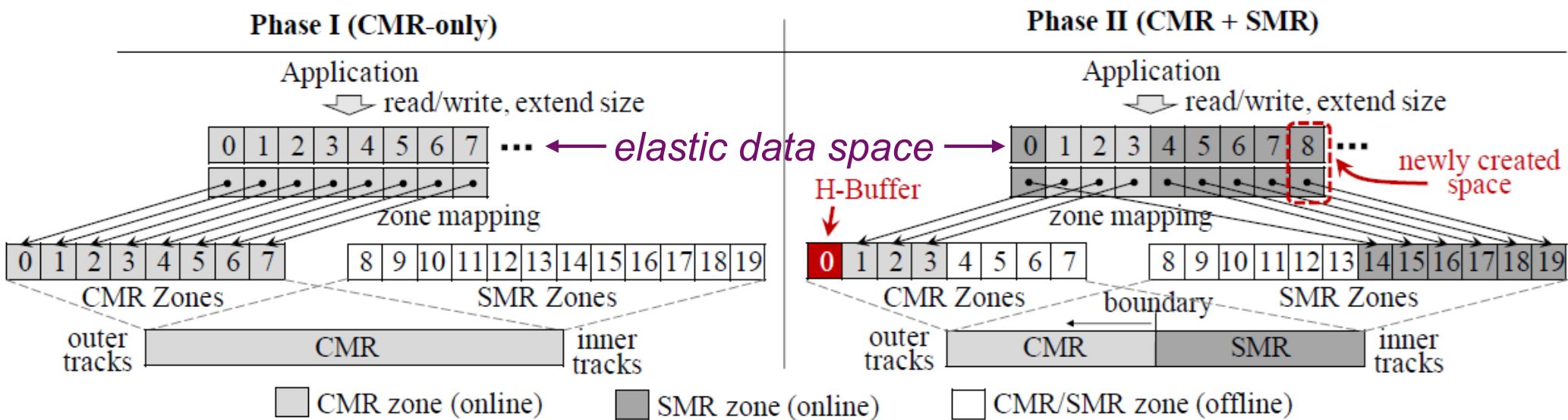
- Google introduces the idea of **Hybrid SMR** recently.
  - A single H-SMR drive can have **both CMR and SMR areas**.
  - The format can be changed from one type to the other.
  - The goal is to balance the IOPS and the capacity.



- There are still many **challenges**:
  - How to efficiently arrange the format layout and place data?
  - How to reduce SMR update overhead?
  - How to adapt to dynamic workloads?

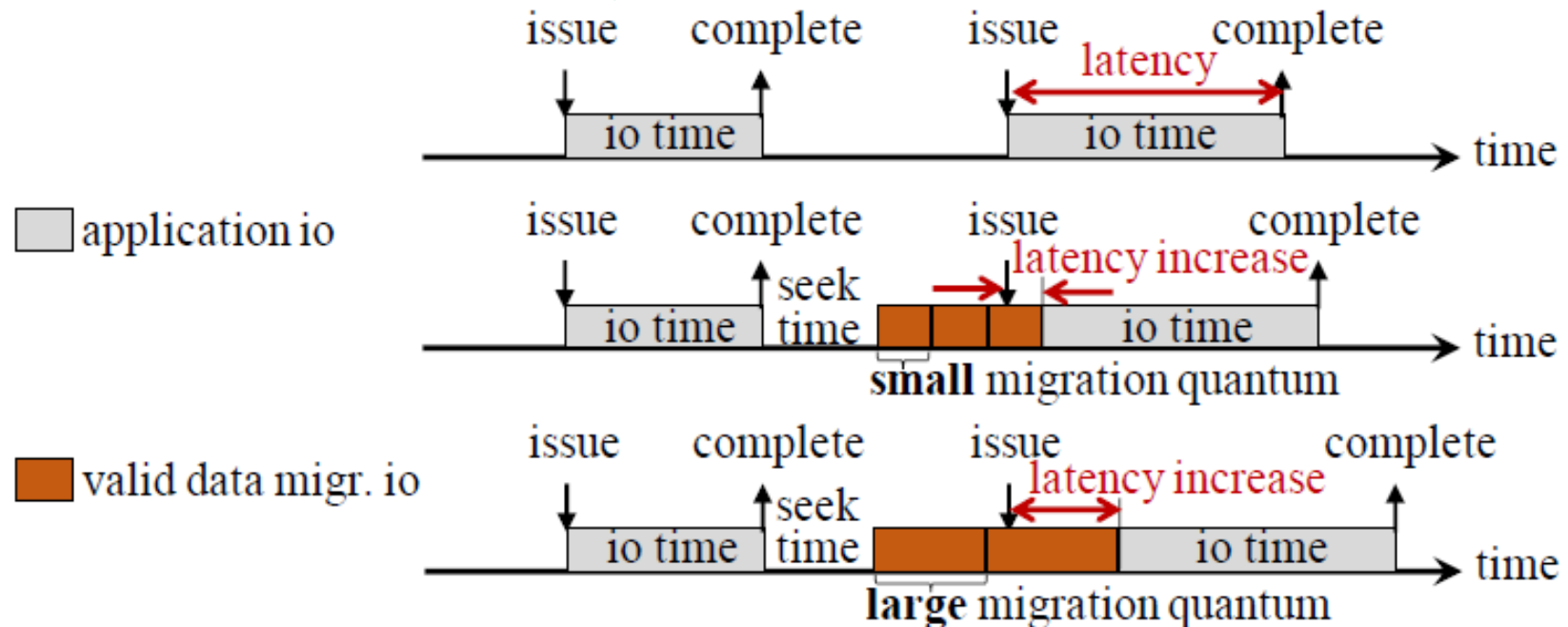
# Case Study of H-SMR: ZoneAlloy

- **ZoneAlloy** hides the H-SMR details and presents upper layer applications with an **elastic data space**.
  - **Elastic Data Space**: an address space with **extendable** size.
  - It can be implemented at the **block driver layer** or **firmware**.
- **ZoneAlloy** adopts a **two-phase** elastic allocation:
  - Phase I: Initially allocating CMR space only
  - Phase II: Converting from CMR to SMR as necessary
    - Through a **quantized migration** to control the impact.



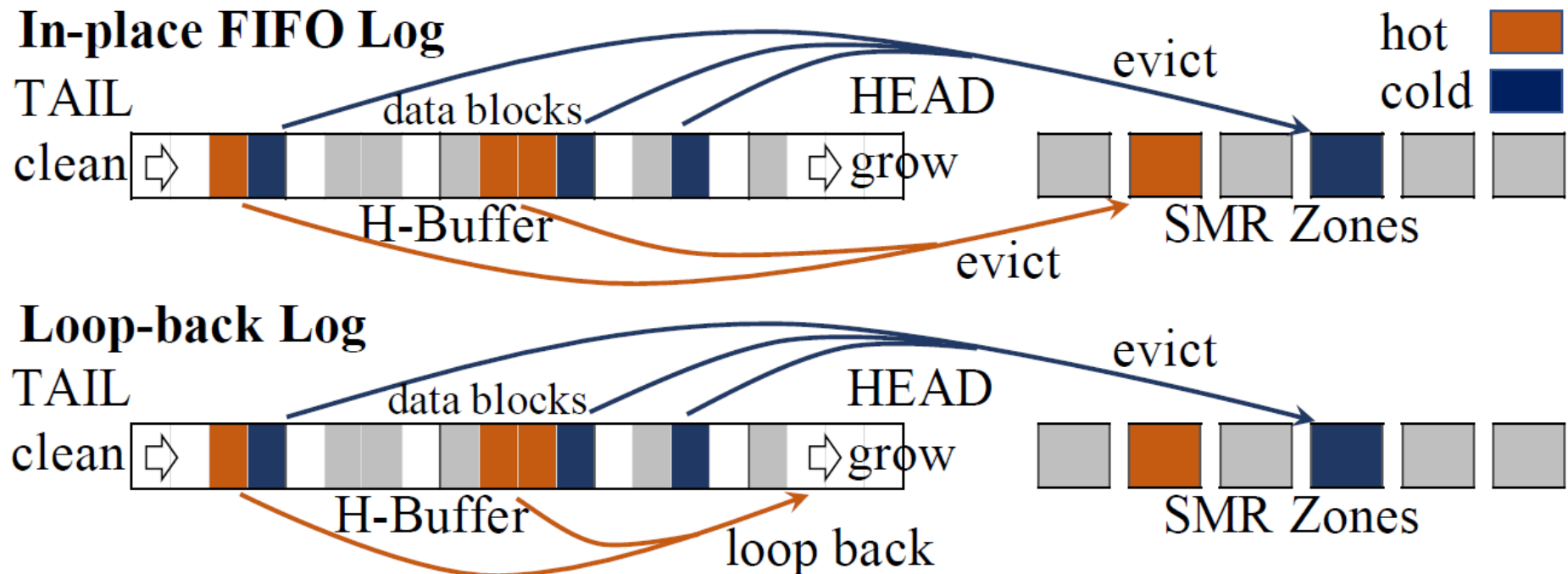
# ZoneAlloy: Quantized Migration

- Conversion is **time consuming** and **intrusive**.
  - The application I/O will be **delayed by seconds**, or even **minutes**, depending on the size of the requested space.
  - There is a trade-off between the performance of the application I/O and the conversion efficiency.
- Performing the migration in the unit of **migration quantum** which can be **decided by applications**.



# ZoneAlloy: Host-controlled Buffer

- Updating an SMR zone directly using **RMW** still introduces significant **performance overhead**.
- A small CMR buffer (called **H-Buffer**) can **accumulate** multiple updates and **migrate** them to SMR **in a batch**.
  - **In-place FIFO Log** organizes the redirected data in a log.
  - **Loop-back Log** keeps hot data in log without evicting them.



# Summary

- Traditional Hard Disk Drive
  - Why and How
  - Development Bottleneck
- New Magnetic Recording Technologies
- Shingled Magnetic Recording (SMR)
  - Basics and Inherent Challenges
  - General Solution: Persistent Cache
  - Various SMR Drive Models and Designs
    - Drive-Managed SMR (DM-SMR)
    - Host-Aware SMR (HA-SMR)
    - Host-Managed SMR (HM-SMR)
    - Hybrid SMR

